
UNIT 14 QUALITATIVE INDEPENDENT VARIABLES IN OLS MODELS *

Structure

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14.0 OBJECTIVES

After going through this Unit you will be able to

- Apply Chow test for structural stability of a regression model;
- explain the nature of dummy variables;
- use dummy variables in regression models;
- test for structural stability in dummy variable models; and
- pool cross-sectional and time-series data by using dummy variables.

14.1 INTRODUCTION

In regression models considered so far, we assumed that there is no change in parameters within a sample. There could be situations, however, where there is a change (or shift) in a parameter within the sample. For example, assume that industrial output of India depends upon inputs such as capital and labour apart from technology and policy environment. After the economic liberalization in 1991, there is a major shift in the economic environment in India - there is an improvement in the level of technology and the level of competition. In short, we can say that there is a structural change in the Indian economy after economic liberalization. In such circumstances we cannot assume that the parameters

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remain the same for the pre-liberalisation period and the post-liberalisation period. A question before us is: How do we represent structural change in a regression model?

Let us consider another scenario. We assumed that the explanatory variables are numerical or quantitative in nature. This may not always be the case. There can be instances where the dependent variable (say, monthly earning) depends upon certain qualitative variable (say, gender). Can we include a qualitative variable such as gender as explanatory variable in a regression model? Yes, we can. A qualitative variable can be represented by a *dummy variable*, which assumes numerical values as we will see later in this Unit.

The dummy variables are also known as *binary variables*, *indicator variables*, *categorical variables*, and *dichotomous variables*. In this unit we consider the role of qualitative explanatory variables in regression analysis. We include dummy variables in regression models and show that it is a flexible tool for handling many interesting problems.

14.2 CHOW TEST FOR STRUCTURAL STABILITY

We begin with the Chow test for structural stability of a regression model. A structural change (also called structural break) occurs if the parameters underlying a relationship differ from one subset of the data to another. There may be several relevant subsets of data with the possibility of several structural breaks in a sample. For the moment we consider two subsets of n_1 and n_2 observations so that sample size is $n = n_1 + n_2$. Suppose, for example, we wish to investigate whether savings rate differs between rich households and poor households. Suppose we have n_1 poor households and n_2 rich households. An option before us is to estimate separate regression equations for the poor and the rich households. By comparing the estimated coefficients of both the regression models we can find out whether there is difference in savings rate of poor and rich households. Alternatively, we can estimate the regression model for the poor households and predict the savings rate for the rich households (or, *vice versa*). It is however not clear which choice should be made, and the two procedures might well yield different answers! Chow test gives concrete idea on the structural stability of a model.

Let us consider regression models for two groups as follows:

Rich households:

$$Y_t = \alpha_1 + \alpha_2 X_t + \varepsilon_{1t} \quad t = 1, 2, \dots, n_1 \quad \dots (14.1)$$

Poor households:

$$Y_t = \beta_1 + \beta_2 X_t + \varepsilon_{2t} \quad t = 1, 2, \dots, n_2 \quad \dots (14.2)$$

where Y is total savings of a household, X is personal income, the ε 's are the error terms in the two equations, n_1 and n_2 are the number of observations in the two sub-samples.

A structural change may mean that the two intercepts are different ($\alpha_1 \neq \beta_1$), or the two slopes are different ($\alpha_2 \neq \beta_2$), or both the intercept and the slopes are different, or any other suitable combination of the parameters. If there is no structural change (i.e., there is structural stability), we can combine all the n_1 and n_2 observations and estimate one savings function as

$$Y_t = \lambda_1 + \lambda_2 X_t + \varepsilon_t \quad \dots (14.3)$$

There are two basic assumptions underlying the Chow test.

- (a) $\varepsilon_{1t} \sim N(0, \sigma^2)$ and $\varepsilon_{2t} \sim N(0, \sigma^2)$, i.e., the two error terms are normally distributed with the same variance, and
- (b) ε_{1t} and ε_{2t} are independently distributed.

The Chow test proceeds as follows: We assume that regression equations for both the groups are different. This is our null hypothesis. Therefore, both the equations (14.1) and (14.2) need to be estimated separately. This is our unrestricted model. On the other hand, we restrict the above scenario and combine both the groups. We estimate equation (14.3) – this is our restricted model. Thus, the alternate hypothesis is our restricted model. The steps for carrying out the Chow test is given below.

- (i) Combine all the n_1 and n_2 observations and estimate regression model (14.3). Obtain restricted residual sum of squares, say RSS_R . The degree of freedom for this model is $(n_1 + n_2 - k)$, where k is the number of parameters in the model.
- (ii) Estimate regression models (14.1) and (14.2) individually. Obtain their residual sum of squares. Let us denote it by RSS_1 and RSS_2 respectively. The degrees of freedom for regression model (14.1) is $(n_1 - k)$ and that for regression model (14.2) is $(n_2 - k)$.
- (iii) Add these two residual sums of squares ($RSS_1 + RSS_2$) and obtain RSS_{UR} (unrestricted residual sum of squares). The degrees of freedom for RSS_{UR} is $(n_1 + n_2 - 2k)$.
- (iv) Apply the following formula to obtain the F-ratio:

$$F = \frac{(RSS_R - RSS_{UR})/k}{RSS_{UR}/(n_1 + n_2 - 2k)} \quad \dots (14.4)$$

which follows the F distribution with $DF = (k, n_1 + n_2 - 2k)$. If the F computed from (14.4) exceeds the critical value at the chosen level of confidence level, we reject the hypothesis that the regressions (14.1) and (14.2) are the same. In other words, we reject the hypothesis of structural stability.

Example 14.1

Let us consider the following hypothetical data on personal savings and personal income by poor and rich households (in Rupees thousand). We want to test whether the savings function is the same in both the groups.

Poor Households			Rich Households		
Sl. No.	Savings	Income	Sl. No.	Savings	Income
1	0.36	8.8	1	0.59	15.5
2	0.21	9.4	2	0.90	16.7
3	0.08	10.0	3	0.95	17.7
4	0.20	10.6	4	0.82	18.6
5	0.10	11.0	5	1.04	19.7
6	0.12	11.9	6	1.53	21.1
7	0.41	12.7	7	1.94	22.8
8	0.50	13.5	8	1.75	23.9
9	0.43	14.3	9	1.99	25.2

In the above data, we have $n_1 = n_2 = 9$.

For group 1, poor households

$$\hat{Y}_t = -0.2623 + 0.0470 X_t$$

(0.3054) (0.0266)

$$t = (-0.8719) \quad (1.7700)$$

$$R^2 = 0.3092, \quad \text{RSS}_1 = 0.1396, \quad DF = 7$$

For group 2, rich households:

$$\hat{Y}_t = -1.7502 + 0.1504 X_t$$

(0.3576) (0.0175)

$$t = (-4.8948) \quad (8.5749)$$

$$R^2 = 0.9131, \quad \text{RSS}_2 = 0.1931, \quad DF = 7$$

For combined data (restricted model):

$$\hat{Y}_t = -1.0821 + 0.1178 X_t$$

(0.1452) (0.0088)

$$t = (-7.4548) \quad (13.4316)$$

$$R^2 = 0.9185, \quad \text{RSS}_R = 0.5722, \quad DF = (9 + 9 - 2) = 16$$

Let us plug in the above data into the Chow test formula given at (14.4).

We have $RSS_{UR} = RSS_1 + RSS_2 = 0.1396 + 0.1931 = 0.3327$

$$F = \frac{(RSS_R - RSS_{UR})/k}{RSS_{UR}/(n_1 + n_2 - 2k)} = \frac{(0.5722 - 0.3327)/2}{0.3327/14} = \frac{0.11975}{0.0238} = 5.04$$

At the 5 percent level of significance, the critical value of $F_{2,14} = 3.74$. Since the observed F value of 5.04 exceeds this critical value, we reject the null hypothesis that the savings function in the two time periods is the same.

On the basis of the Chow test we can draw the conclusion that the savings functions in the two groups are different; but what is the reason for this difference? Is it due to the difference in the intercept values, or the slope values, or in both? The Chow test does not provide any clue on this. The dummy variable model discussed below helps us in finding an answer to this question.

14.3 THE NATURE OF DUMMY VARIABLES

In regression analysis the dependent variable is frequently influenced not only by variables that can be readily quantified on some well-defined scale (e.g., income, output, prices, costs, weights, etc.), but also by other variables that are essentially qualitative in nature (e.g., marital status, gender, religion, caste, etc.). For example, holding all other factors constant, IIT graduates are found to earn more than their counterparts from the regional engineering colleges in India. Similarly, studies in the US have reported that female college teachers earn less than their male counterparts. Whatever may be the reason for this disparity, qualitative variables like gender, institution of education, etc. do influence the dependent variable and should be included among the independent variables.

Since qualitative variables usually indicate the presence or absence of some attribute or quality, such as rural or urban, male or female, married or unmarried, etc. We can quantify such attributes by constructing artificial variables that take value 1 or 0, 1 indicating the presence (or possession) of a particular attribute and 0 the absence of it and vice versa. For example, 1 may indicate that the person is a male and 0 may indicate that the person is female; or 1 may indicate that a person is educated and 0 that he/she is not educated and so on. Such variables which assume values 0 and 1 are called Dummy Variables.

Like the quantitative variables, the dummy variables can be used in regression analysis very easily. There can be two cases: (i) a regression model may contain only dummy explanatory variables. This type of regression models is called *analysis of variance (ANOVA) models*. (ii) a regression model may contain both quantitative variables and dummy variables. This type of regression models is called the *analysis of variance (ANOVA) models*.

14.3.1 ANOVA Models

The following model is an example of ANOVA model

$$Y_i = \alpha_1 + \alpha_2 D_i + u_i \quad \dots (14.5)$$

where, Y_i = annual starting salary of a school-teacher

$$D_i = \begin{cases} 1 & \text{if male teacher} \\ 0 & \text{otherwise (i.e., female teacher)} \end{cases}$$

Model (14.5) is similar to an ordinary two variable regression model. The only difference is the use of a qualitative or dummy variable D instead of quantitative explanatory variable X . (From now on in the present unit we shall be using D to denote dummy variable). Assuming all the other factors such as age, years of experience, etc. to be constant, the model (14.5) may enable us to find out whether gender (i.e., being a male or female) makes any difference in a school teacher's salary. In other words, it would enable us to find out whether a male school teacher's salary is different from that of a female school teacher having the same qualifications and years of experience.

Inclusion of the disturbance term u_i in the model (14.5) satisfies the usual assumptions of the classical linear regression model (CLRM). We obtain from (14.5) that the mean salary of a female school-teacher is:

$$E(Y_i | D_i = 0) = \alpha_1 + \alpha_2(0) = \alpha_1 \quad \dots (14.6)$$

and the mean salary of a male school-teacher is:

$$E(Y_i | D_i = 1) = \alpha_1 + \alpha_2(1) = \alpha_1 + \alpha_2 \quad \dots (14.7)$$

In the above model the intercept term α_1 gives the mean annual salary of a female school-teacher. The slope coefficient α_2 tells us how much the mean salary of a male school-teacher differs from that of his female counterpart with $(\alpha_1 + \alpha_2)$ reflecting the mean annual salary of a male school-teacher.

We can also test for the following hypothesis: Is there discrimination in accordance to the gender of an individual while determining the salary of school-teachers. Here, we run by running OLS on the regression equation (14.5) and find out on the basis of t -test whether the estimated α_2 is statistically significant or not.

We will demonstrate the above with the help of a hypothetical example. Let us consider Table 14.1 which gives hypothetical data of annual salary of school-teachers.

Table 14.1: Annual Salary of School-teachers

Salary(Y) (Rs. 000)	Gender Dummy (D) (Male =1, Female = 0)
22.0	1
19.0	0
21.2	1
17.5	0
17.0	0
18.0	0
21.7	1
18.5	0
21.0	1
20.5	1

The OLS results corresponding to the regression model (14.5) are as follows:

$$Y_i = 18.00 + 3.28 D_i \quad \dots (14.8)$$

Standard error (0.32) (0.44)

t-statistic (57.74) (7.44)

$$R^2 = 0.8737$$

From (14.8) we see that the estimated mean salary of female school-teachers ($\hat{\alpha}_1$) is Rs.18,000 and that of male school-teachers ($\hat{\alpha}_1 + \hat{\alpha}_2$) is Rs.21,280.

From the calculated t-statistic in regression model (14.8), it is easy to verify that $\hat{\alpha}_2$ is statistically significant (i.e., significantly different from zero), suggesting a difference in starting salaries of male and female school-teachers. If we represent the regression model (14.8) a diagram, we obtain a step-function of the type given in Fig. 14.1.

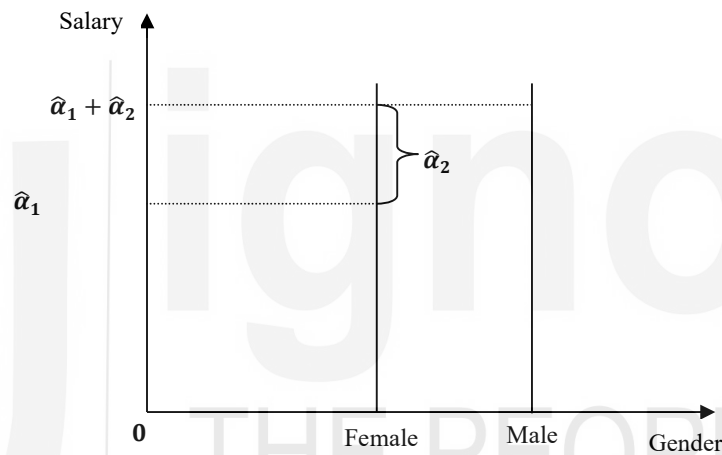


Fig. 14.1: Salary of Male and Female Teachers (Rs. '000)

ANOVA models such as those in (14.8), although frequently used in fields such as sociology, education, market research, etc. are not common in economics. Regression models in most economic research consist of explanatory variables that are both quantitative and qualitative in nature. Such type of regression models which contain both qualitative and quantitative explanatory variables are known as *analysis of covariance (ANCOVA) models*.

14.3.2 ANCOVA Models

As mentioned earlier, a regression model can have both qualitative (dummy) variables and quantitative variables. Such models are called ANCOVA models. Let us now consider the following as an example of the ANCOVA model:

$$Y_i = \alpha + \beta D_i + \gamma X_i + u_i \quad \dots (14.9)$$

where, Y_i = annual salary of a school-teacher

X_i = years of teaching experience

D_i = 1 if male teacher

0 otherwise (i.e., female teacher)

Regression model (14.9) is similar to (14.5) except that it contains a quantitative variable (years of teaching, X_i) in addition to the qualitative variable (gender of the teacher, D_i). The qualitative variable has two categories, namely, male and female. Thus, the model has two explanatory variables: one qualitative and one quantitative.

What does model (14.9) mean/imply? Assuming, as usual, $E(u_i) = 0$, we see that the mean salary of a female school-teacher is

$$E(Y_i | D_i = 0) = \alpha + \gamma X_i \quad \dots (14.10)$$

and the mean salary of male school-teacher is

$$E(Y_i | D_i = 1) = (\alpha + \beta) + \gamma X_i \quad \dots (14.11)$$

From the model (14.9) we see that the male and female school-teachers' salary function in relation to the years of experience have the same slope (γ) but different intercepts. In other words, in the model, it is assumed that the level of mean salary of male school-teacher is different from that of his female counterpart but the rate of change in the mean salary by years of experience (represented by the slope term γ) is the same for both male and female school-teachers. Geometrically we can represent models (14.6) and (14.7) as shown in Fig. 14.2 (in the figure α is assumed to be positive, i.e., $\alpha > 0$).

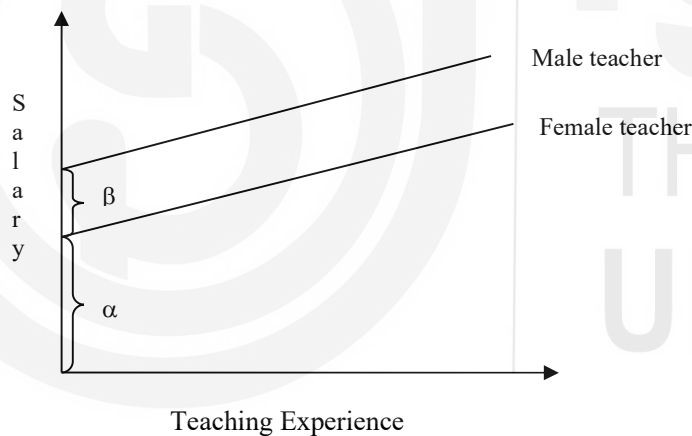


Fig. 14.2: Annual Salary of Male and Female Teachers

Just as in regression (14.8), here also we can use the t -test to test for the hypothesis that male and female school-teachers have the same mean annual salary.

Before proceeding further, it is essential to discuss some of the important features of the dummy variables:

- (i) In dummy variable models discussed so far, we have introduced only one dummy variable, D_i to distinguish between two categories. For example, male and female with $D_i = 1$ denoting male and $D_i = 0$ denoting female. Now what happens if instead of one dummy variable two dummy variables

D_{1i} and D_{2i} are introduced in the model, one each for male and female? Model (14.9) can now be written as

$$Y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + \gamma X_i + u_i \quad \dots (14.12)$$

where, Y_i and X_i are as defined before, and

$$D_{1i} = 1 \quad \text{if a male teacher}$$

$$0 \quad \text{otherwise}$$

$$D_{2i} = 1 \quad \text{if a female teacher}$$

$$0 \quad \text{otherwise}$$

Due to perfect collinearity between D_1 and D_2 (i.e., perfect linear relationship) the regression model (14.12) cannot be estimated. This can be explained with the help of the data given in Table 14.2.

Table 14.2: Annual Salary of School Teachers

		Intercept	D_1	D_2	X
Male	Y_1	1	1	0	X_1
Male	Y_2	1	1	0	X_2
Female	Y_3	1	0	1	X_3
Male	Y_4	1	1	0	X_4
Female	Y_5	1	0	1	X_5

From Table 14.2 it is easy to verify that D_1 and D_2 are perfectly collinear as $D_1 = (1 - D_2)$ or $D_2 = (1 - D_1)$. We know from Unit 10 that in the case of perfect multicollinearity, it is not possible to estimate the various parameters.

There are, however, several ways of resolving this problem but the simplest one is by using dummy variables as we had done in model (14.5). We should use only one dummy variable if there are two categories of a qualitative variable.

Rule of Thumb: If a qualitative variable has 'm' categories, introduce '(m - 1)' dummy variables.

Thus, if a qualitative variable has 4 characteristics, we introduce only 3 dummy variables. If this rule is not followed, we will fall into what is known as the *dummy variable trap*, i.e., a situation of perfect multicollinearity.

- (ii) The assignment of values 0 and 1 to two categories such as rural and urban, or educated and uneducated, etc., is arbitrary. For example, in our model (14.5) instead of assigning 1 to male teacher and 0 to female teacher we could have assigned the value 1 to female teacher and 0 to male teacher

(and the coefficients would change accordingly). In such a case, how do we interpret the results? While interpreting the results of the models that use dummy variables, it is critical to know how the values 1 and 0 are assigned. The category that is assigned value 0 is referred to as the ‘base category’ or ‘benchmark category’ and all the comparisons are made with reference to this category. In model (14.5) female school-teacher, which is assigned value 0, is the base or benchmark category.

- (iii) The coefficient attached to the dummy variable (for example β in model 14.5) is referred to as the differential intercept coefficient because it tells by how much the value of the intercept term of the category that receives value 1 differs from that of the base category.

Check Your Progress 1

- 1) Under what circumstances is the use of dummy variable suggested? Give some practical examples where dummy variables can be fitted.

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- 2) Suppose an explanatory variable is divided into 4 categories and you plan to fit 4 dummies in the regression equation. Is it possible to estimate the equation?

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14.4 USE OF MORE THAN ONE DUMMY VARIABLE

We can extend the analysis in the previous section to handle more than one qualitative variable with ease. Consider once again the example of salary of school-teacher which was used in the previous section. In this section we assume that the salary of a school-teacher to be dependent not only on the years of experience and gender of the teacher but also on the type of education of the teacher. In the present example it is assumed that the type of education has two categories, convent educated and non-convent educated. Incorporating the variable type of education of the teacher in regression model (14.5) we can write it as:

$$Y_i = \alpha + \beta D_{1i} + \gamma D_{2i} + \delta X_i + u_i \quad \dots (14.13)$$

where, Y_i and X_i are as defined before and

$$\begin{aligned}
 D_{1i} &= 1 && \text{if a male teacher} \\
 &= 0 && \text{otherwise (i.e., female teacher)} \\
 D_{2i} &= 1 && \text{if convent educated} \\
 &= 0 && \text{otherwise (i.e., non-convent educated)}
 \end{aligned}$$

In the above model (14.13) we have used two dummy variables one each for the two qualitative explanatory variables ‘gender of the teacher’, and ‘type of education of the teacher’. The base or the benchmark category in the above model is ‘non-convent educated female school-teacher’.

Assuming $E(u_i) = 0$, from equation (14.13), we can have the following:

Mean salary of a non-convent educated female school-teacher:

$$E(Y_i | D_1 = 0, D_2 = 0, X_i) = \alpha + \delta X_i \quad \dots (14.14)$$

Mean salary of non-convent educated male school-teacher:

$$E(Y_i | D_1 = 1, D_2 = 0, X_i) = (\alpha + \beta) + \delta X_i \quad \dots (14.15)$$

Mean salary of convent educated female school-teacher:

$$E(Y_i | D_1 = 0, D_2 = 1, X_i) = (\alpha + \gamma) + \delta X_i \quad \dots (14.16)$$

Mean salary of convent educated male school-teacher:

$$E(Y_i | D_1 = 1, D_2 = 1, X_i) = (\alpha + \beta + \gamma) + \delta X_i \quad \dots (14.17)$$

A variety of hypotheses can be tested by the Ordinary Least Squares (OLS) estimation of the model (14.13). For example, we can test for the significance of the differential intercept terms β or γ , or both β and γ thereby determining which of the possibilities exists without running the regression separately for each combination of gender and type of education.

In a similar fashion we can extend the model to include more than one quantitative and more than two qualitative variables. However, we must keep in mind that *the number of dummies for each of the qualitative variable should be one less than the number of categories of that variable.*

14.5 TESTING FOR STRUCTURAL STABILITY THROUGH DUMMY VARIABLES

In the regression models that have been discussed so far in the present unit we have considered that the qualitative variables affect the intercept term only but not the slope coefficient. But what happens if the slope coefficients are also affected by the qualitative variables? In such situations, testing the differences in the intercepts alone will be of little significance. Therefore, we need to look for a methodology that will identify whether the differences in two or more regressions are due to differences in the intercept, or slopes or both slope and intercept. In order to understand this problem, let us consider the following example. Suppose we are interested in estimating a simple savings function that relates domestic household savings (S) with the gross domestic product (Y) of India for the period

1980-81 to 2002-03. The relevant data is given in Table 10.3 below. One way of proceeding is to simply run an OLS regression of S on Y for the entire period 1980-81 to 2002-03 if the relation between savings and GDP do not change over the entire period. But it may not be so. India, in 1991 introduced a series of economic reforms thereby bringing in substantial change in its economic system. The introduction of economic reforms would have also influenced considerably the savings-income relationship. Thus, our objective is now to check whether the savings-income relationship has undergone a structural change between the two time periods or not. By structural change we mean that the parameters of the savings function have changed. Let us assume the model

$$S_t = A_1 + A_2 Y_t + u_{1t} \quad \dots (14.18)$$

for the pre-reforms period; and

$$S_t = B_1 + B_2 Y_t + u_{2t} \quad \dots (14.19)$$

for the post-reforms period

where, S_t = household savings in period t, (t = 1980-81 to 2002-03)

Y_t = gross domestic product in period t, (t = 1980-81 to 2002-03)

u = the error term

The Chow test would tell us whether there was a structural change in the saving-income relationship over the concerned period. However, it will not tell us whether the difference in the two regression models (14.18) and (14.19) is in their intercept values or in the slope values or in both. Comparing the two models (14.18) and (14.19) we see that there are four possibilities: (these possibilities are illustrated in Fig.14.3)

- $A_1=B_1$, and $A_2=B_2$; i.e., the two regressions are identical. This is the case of *coincident regression*. (refer to Fig.14.3a)
- $A_1 \neq B_1$, but $A_2=B_2$; i.e., the two regressions differ only in their location or the intercepts. This is a case of *parallel regression*. (refer to Fig.14.3b)
- $A_1=B_1$, but $A_2 \neq B_2$; i.e., the two regressions have same intercept term but different slopes. This is a case of *concurrent regression*. (refer to Fig.14.3c) and
- $A_1 \neq B_1$, and $A_2 \neq B_2$; i.e., the two regressions are completely different. This is a case of *dissimilar regression*. (refer to Fig.14.3d)

From the data on household savings and gross domestic product for India given in Table 14.2 we can run the regressions (14.18) and (14.19) and apply the Chow test to see whether the savings function has undergone a structural change between the two time periods. (Students are advised to apply the Chow Test and find out the result as an assignment)

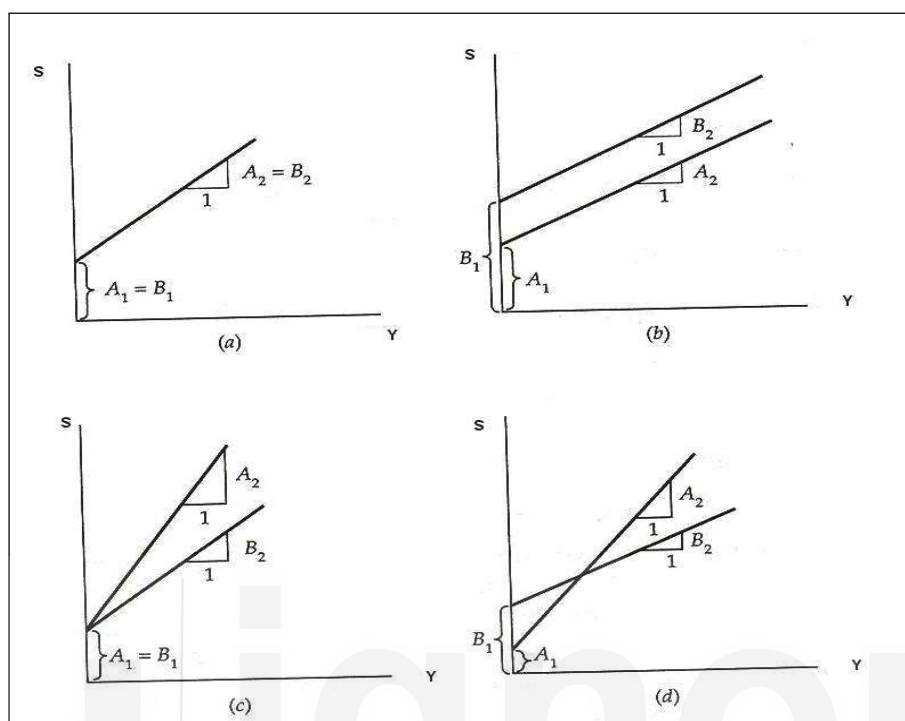


Fig. 14.3: Possible Differences in the Two Regression Models

Table 14.3: Domestic Savings and GDP in India, 1980-81 to 2002-03

(at constant prices) **(Rs. crore)**

Year	Domestic Savings (S)	Gross Domestic Product (Y)	Year	Domestic Savings (S)	Gross Domestic Product (Y)
1980-81	27136.0	401128.0	1992-93	162906.0	737791.6
1981-82	31355.0	425072.8	1993-94	193621.3	781345.0
1982-83	34368.0	438079.5	1994-95	251463.4	838031.0
1983-84	38587.0	471742.2	1995-96	298747.3	899563.0
1984-85	46063.0	492077.3	1996-97	317260.7	970082.0
1985-86	54167.0	513990.0	1997-98	352178.0	1016595.0
1986-87	58951.0	536256.7	1998-99	374659.0	1082748.0
1987-88	72908.0	556777.8	1999-00	468681.0	1148368.0
1988-89	87913.0	615098.4	2000-01	495986.0	1198592.0
1989-90	106979.0	656331.2	2001-02	535185.0	1267833.0
1990-91	131340.0	692871.5	2002-03	597697.0	1318321.0
1991-92	143908.0	701863.2			

An alternative way of testing whether the savings function has undergone a structural change or not is the use of *dummy variables*. Let us see how the dummy variables can be used to handle the problem of structural change.

Let us write the savings function as

$$S_t = a + bD_t + cY_t + d(D_tY_t) + u_t \quad \dots (14.20)$$

where, S_t = household savings in period t

Y_t = GDP in period t

$D_t = 1$ for observation in the pre-reform period

= 0 otherwise, i.e., for observation in the post-reform period

In model (14.20) b is the *differential intercept* and d is the *differential slope coefficient* indicating how much the slope coefficient of the pre-reform periods savings function differs from the slope coefficient of the savings function in the post reform period. Just as the introduction of an additive dummy enables us to distinguish between the intercepts of the two periods, the introduction of multiplicative dummy enables us to differentiate between the slope coefficients in the two periods.

Let us see the implications of the regression model (14.20). Assuming $E(u_t) = 0$, we obtain

$$E(S_t|D_t = 0, Y_t) = \alpha + cY_t \quad \dots (14.21)$$

$$E(S_t|D_t = 1, Y_t) = (\alpha + b) + (c + d)Y_t \quad \dots (14.22)$$

The above two equations are the mean savings function for the pre-reforms period and the post-reforms period respectively as represented by model (14.18) and (14.19) with $a=A_1$, $c=A_2$, and $(a+b)=B_1, (c+d)=B_2$. Thus, with the help of dummy variables a single regression can easily be used to obtain two sub-period regressions.

Now estimating the regression (14.20) using the savings and income data given in Table 14.3 we obtain

$$\hat{S}_t = -133690.9 - 228328.9 D_t + 0.375 Y_t + 0.3385 D_t Y_t \quad \dots (14.23)$$

$$SE \quad (21259.0) \quad (30775.09) \quad (0.0386) \quad (0.0441)$$

$$t\text{-stat} \quad (-6.289) \quad (-7.429) \quad (9.717) \quad (7.674)$$

$$R^2 = 0.9954$$

We can see from the estimated regression model (14.23) that both the differential intercept and the differential slope coefficients are statistically significant. It implies that the regression models or the savings-income relationship for both the periods are different. Both the regression models can be diagrammatically represented by Fig.14.3d. Substituting for D_t the values 0 and 1 we obtain two separate equations as given below.

For the pre-reforms period, when $D_t = 0$, we have

$$\hat{S}_t = -133690.9 - 228328.9 (0) + 0.375 Y_t + 0.3385(0) Y_t$$

$$\text{Or, } \hat{S}_t = -133690.9 + 0.375 Y_t \quad \dots (14.24)$$

For the post-reforms period we have

$$\begin{aligned}\hat{S}_t &= -133690.9 - 228328.9(1) + 0.375 Y_t + 0.3385(1) Y_t \\ \text{Or, } \hat{S}_t &= -133690.9 - 228328.9(1) + 0.375 Y_t + 0.3385(1) Y_t \\ &= -362319.8 + 0.7135 Y_t \quad \dots (14.25)\end{aligned}$$

(Note: You can see that these regression models are the same as those obtained from the Chow test procedure).

Thus, from the above discussion we can infer that the dummy variable technique has certain advantages over the Chow test.

- The dummy variable technique requires running only a single equation while for the Chow test one has to run a number of regressions, one each for each of the periods concerned.
- The Chow test does not tell us whether the intercept or the slope coefficient or both are different in the two sub-samples. In fact, we cannot tell by the Chow test which of the four possibilities presented in Fig.14.3 exists. In this respect the use of dummy variables approach has a definite advantage. It not only tells whether the two regressions are different but also pinpoints the source of their difference – whether it is because of the intercept or the slope or both the coefficients.
- The use of a single regression equation as in the case of dummy variable approach would mean higher degrees of freedom compared to more than one regression model (in the case of Chow test). Thus, the dummy variable technique improves the relative precision of the estimated parameters.

14.6 USE OF DUMMY VARIABLES IN SEASONAL ANALYSIS

Many economic time series based on monthly or quarterly data exhibit seasonal patterns. For example, the sale of woollens during the winter months, sales of departmental stores during the festive season, sale of soft drinks in the summer months, sale of refrigerators and air-conditioners in the summer season etc are some of the examples of time series that show seasonal pattern. It is often desirable to eliminate the seasonal component of such times-series. The process of removing/eliminating the seasonal factor is known as de-seasonalisation or seasonal adjustment. The adjusted time series thus obtained is called the seasonally-adjusted, or the de-seasonalised times series.

There are many ways of deseasonalising a time series. Dummy variables can also be effectively used for deseasonalising a time series. Let us deal with the use of dummy variable method in deseasonalisation of time series. Let us consider the sales of a departmental store. Table 14.4 gives data on the quarterly sales and profits of a departmental store in New Delhi during the period 2021-22 and 2024-25.

Table 14.4: Sales and Profits of a Departmental Store (Rs. Lakhs)

Year	Quarter	Profit (Y)	Sales (X)
2021-22	I	16.63	132.1
	II	19.41	136.5
	III	24.01	177.7
	IV	19.91	157.5
2022-23	I	17.55	150.7
	II	20.90	152.4
	III	25.61	199.8
	IV	21.97	173.0
2023-24	I	19.46	167.4
	II	22.14	179.1
	III	27.42	227.0
	IV	22.72	191.6
2024-25	I	21.42	185.0
	II	25.41	187.3
	III	32.07	261.5
	IV	25.49	219.2

The profit function of the departmental store is dependent on the sales. Let us represent the profit function as

$$Y_t = A_1 + A_2D_{2t} + A_3D_{3t} + A_4D_{4t} + A_5X_t + u_t \quad \dots (14.26)$$

where, Y_t = profit of the store in period t

X_t = sales in period t

The seasonal dummies are defined as

$$D_{2t} = 1 \text{ if the profit lies in the second quarter}$$

= 0 otherwise

D_{3t} = 1 if the profit lies in the third quarter
= 0 otherwise

D_{4t} = 1 if the profit lies in the fourth quarter
= 0 otherwise

In the above model we have, through the four quarters, incorporated seasonal variation. It is assumed that the qualitative variable 'season' has four categories. Therefore, it requires the use of three dummy variables (D_{2t}, D_{3t}, D_{4t}) in the regression model. Note that the first quarter has been taken as the base or benchmark quarter. We have used intercept dummy variables only. Each of these differential intercepts tells us the extent by which the mean value of Y (i.e., the mean profit) differs in each quarter compared to the base quarter. Using the data in Table 14.4, the regression results of equation (14.26) are as follows:

$$\hat{Y}_t = 3.9573 + 2.7314D_2 + 3.1321D_3 + 1.2841D_4 + 0.0932X_t \quad \dots (14.27)$$

SE	(1.195)	(0.491)	(0.642)	(0.525)	(0.007)
t-value	(3.313)	(5.568)	(4.880)	(2.445)	(12.95)

$R^2 = 0.977$

We observe from the t-values that all the differential intercepts are statistically significant. It implies that the average profits differ across each of the four quarters. It shows that the average profit is the maximum in the third quarter (October-December) when the sales are the highest on account of the festive season in the country.

In the regression model (14.27) assumed that the seasonal variations affect the intercept only and not the slope parameter. But it may not be so in reality. To find out whether the seasonal variation have effects on the intercept or the slope (or both), we use the technique of differential intercept and differential slope coefficient discussed in the previous section (model 14.16). Applying the method we can rewrite model (14.26) as shown by regression model (14.28) and test for the significance of differential intercept and differential slope terms.

$$Y_t = A_1 + A_2D_{2t} + A_3D_{3t} + A_4D_{4t} + A_5X_t + A_6(D_{2t}X_t) + A_7(D_{3t}X_t) + A_8(D_{4t}X_t) + u_t \quad \dots (14.28)$$

where, the differential slope coefficients A_6 , A_7 and A_8 tell us the amount by which the slope coefficient of the second quarter, the third quarter and the fourth quarter differ from the base or the first quarter respectively.

14.7 POOLING CROSS SECTION AND TIME SERIES DATA

Let us consider Table 14.5 which gives data on the energy demand (in million tons of oil equivalent) and value of output (in Rs. million) for three sectors of the Indian economy. This is an example of a cross section time series data where we are interested in finding out the relation between energy demand and value of output for three sectors of the economy and for each sector we have data for 18 years from 2005-06 to 2023-24.

There are several ways to study the relationship between energy demand and value of output. First, we can run the following regression models for each sector separately:

Let us consider

$$Y_t = A_1 + A_2 X_t + u_t \quad \dots (14.29)$$

for the agricultural sector;

$$Y_t = B_1 + B_2 X_t + u_t \quad \dots (14.30)$$

for the industrial sector; and so on.

where, Y_t = energy consumption

X_t = sectoral value of output

Using the dummy variable technique as discussed earlier or the Chow test we can find out if the parameters of these demand functions are the same or not.

The second way is to estimate for each year, the cross-sectional regression. In such a case there would be one regression for each of the 18 years giving a total of 18 regressions to be estimated.

The third way is to pool all the 54 observations (18 times series observations for the three sectors) and estimate the following regression

$$Y_{it} = A_1 + A_2 D_{1t} + A_3 D_{2t} + A_4 X_{it} + u_{it} \quad \dots (14.31)$$

where, i stands for i -th sector and t for the t -th time period.

In (14.31) we assumed that only the intercept terms differ across the sector but not the slope terms. You can assume both the slope coefficients and intercept terms to be different across sectors and test of their significance themselves).

Estimation of (14.31) using data in Table 14.5 gives us

$$\hat{Y}_t = 13.133 - 17.426 D_{1t} + 6.696 D_{2t} + 0.0004 Y_t \quad \dots (14.32)$$

Std error (1.482) (1.682) (3.010) (0.0000)

t -statistic (9.195) (-10.356) (2.224) (18.195)

$R^2 = 0.97602$

The above results show that both the dummies are statistically significant. It implies that the three sectors have different intercept terms (As an exercise you

should incorporate slope dummies and test whether they are also significant or not).

Table 14.5: Energy Demand and Value of Output

Year	<i>Agricultural Sector</i>		<i>Industrial Sector</i>		<i>Transport Sector</i>	
	Energy demand	Value of output	Energy demand	Value of output	Energy demand	Value of output
	(mtoe)	(Rs. Mn)	(mtoe)	(Rs. Mn)	(mtoe)	(Rs. Mn)
05-06	5.47	466490	49.63	950820	18.04	249630
06-07	5.67	491390	54.74	1026470	18.45	263940
07-08	6.60	483580	58.32	1064180	19.46	271850
08-09	6.80	535250	61.64	1150020	20.39	283070
09-10	7.80	535430	63.63	1218210	21.09	302310
10-11	8.56	547970	67.71	1274720	22.53	324270
11-12	10.46	533600	72.80	1362240	23.67	344410
12-13	13.86	535290	72.24	1452530	25.55	368640
13-14	15.04	622140	79.13	1586490	27.23	384790
14-15	16.38	632630	83.02	1750530	28.83	410240
15-16	18.81	656530	86.42	1886010	29.09	428940
16-17	21.78	641180	90.34	1875600	30.72	455950
17-18	23.72	680090	93.76	1949940	31.83	477090
18-19	26.62	705130	97.43	2051620	33.35	511310
20-21	29.40	722640	102.17	2241360	35.01	560170
21-22	32.70	703150	101.10	2523590	38.67	623170
22-23	32.82	758880	109.51	2702180	41.65	674410
23-24	34.29	823410	112.24	2817880	43.45	727850

- 1) Consider the data given in Table 14.3 and divide it into two sub-periods 1980-81 to 1991-92 and 1992-93 to 2002-03. Test for structural break in the savings functions using Chow test.

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- 2) Use the data given at Table 14.3 and run the regression given at equation (10.20). Are the differential intercept and differential slope coefficients statistically significant?

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14.8 LET US SUM UP

In many instances, in regression equations, we have explanatory variables which are qualitative in nature. It is difficult to quantify these qualitative variables as they at best can be divided into certain categories. In such cases we use the dummy variable model.

The dummy variable can affect the intercept or slope or both. Accordingly, we take intercept or slope dummies. Remember that dummy variables are used, as are explanatory variables, on the basis of the logic we build up. Thus, behind every regression model there is a theoretical basis.

Dummy variables can be used in seasonal analysis. It also can be used in pooling cross-sectional and time series data.

14.9 KEY WORDS

ANOVA Models: This is similar to the case where only dummy variable is taken as explanatory variable.

ANCOVA Models: This is similar to regression model having both qualitative and quantitative variables as explanatory variables.

Cross-section Data: This refers to data measured over geographically dispersed units at a point of time.

Dummy Variable: A qualitative variable which takes values that can be divided into categories.

Time Series Data: This refers to data measured over a period of time at regular intervals.

14.10 SOME USEFUL BOOKS

Gujarati, D N and D C Porter (2009), *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S. and K. Lahiri (2012), *Introduction to Econometrics*, Fourth Edition, Wiley

Verbeek, M. (2012), “*A Guide to Modern Econometrics*”, Fourth Edition, John Wiley & Sons

Wooldridge, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

14.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Dummy variable is suggested where explanatory variable is qualitative in nature. Go through Section 14.3.
- 2) It will lead to a situation of perfect multicollinearity. Go through Unit 10 to find the consequences.

Check Your Progress 2

- 1) The procedure of Chow test is given in Section 14.2. On that basis test the model.
- 2) Go through Section 14.4 and formulate the model with differential intercept and slope coefficients. Test the model based on t-test of the dummy variable coefficients.

UNIT 15 QUALITATIVE DEPENDENT VARIABLES IN OLS MODELS*

Structure

- 15.0 Objectives
- 15.1 Introduction
- 15.2 Linear Probability Model
 - 15.2.1 Specification of the Model
 - 15.2.2 Limitations of the Model
 - 15.2.3 Interpretation of LPM
- 15.3 Logit Model
 - 15.3.1 Features of the Logit Model
 - 15.3.2 Estimation of Logit Model
 - 15.3.3 Interpretation of Logit Model
- 15.4 Probit Model
 - 15.4.1 Specification of Probit Model
 - 15.4.2 Estimation of Probit Model
 - 15.4.3 Interpretation of Probit Model
- 15.5 Joint Significance in Qualitative Response Regression Models.
- 15.6 Goodness-of-Fit in Logit and Probit Models
- 15.7 Choice between Logit and Probit Models
- 15.8 Let Us Sum Up
- 15.9 Key Words
- 15.10 Some Useful Books
- 15.13 Answers / Hints to Check Your Progress Exercises

15.0 OBJECTIVES

After going through this Unit, you will be in a position to

- explain how qualitative (dummy) variables can be incorporated as dependent variable in a regression model;
- describe the advantages and disadvantages of linear probability model (LPM);

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- explain how logit and probit models can remove the disadvantages of LPM;
- calculate and interpret marginal effects in LPM, Logit model, and Probit model; and
- interpret Stata software generated output for all three models.

15.1 INTRODUCTION

In Unit 14 we discussed the use of qualitative variables (dummy variables) as explanatory variables in a regression model. Such qualitative variables can assume binary values of 0 and 1. A question may be coming to your mind: Can we use a dummy variable as explained variable (or, dependent variable) in a regression model? The answer is: Yes.

In this Unit we will look at regression models where the dependent variable, Y , is a qualitative variable. Such models are called ‘limited dependent variable models’, or dummy dependent variable models. The term ‘limited dependent variable’ was first used by James Tobin in 1958 to denote regression models where the dependent variable is constrained to be non-negative. As you know, nominal variables can assume more than two categories. If the dummy dependent variable has exactly two categories, it is called a binary (or dichotomous) variable. Otherwise, it is called *polychotomous variable* (containing more than two categories). The present Unit is introductory in nature, and we will focus on binary cases where a dummy variable can take only two values, 0 and 1.

If the *dependent* variable of the model is a dummy variable, the usual OLS method will not be applicable. Instead, the method of *maximum likelihood estimation* is used. This is because when the dependent variable is a dummy variable, the objective is to find out the *maximum probability* (hence the name *maximum likelihood*) of occurrence of the given set of data.

Binary choice models are commonly used for micro analysis in social sciences and medical research. There are three types of ‘limited dependent variable models’. These are: (i) Linear Probability Models (LPM), (ii) Logit Models, and (iii) Probit Models. The last two models have quite similar probability density functions, similar estimation procedures, and they are often used interchangeably.

15.2 LINEAR PROBABILITY MODEL

A linear probability model (LPM) is a regression model where the dependent variable (regressand) is a binary (or, dummy) variable, and the independent variables are either dummy or continuous.

15.2.1 Specification of the Model

To fix ideas, let us consider the following linear probability regression model on determinants of house ownership

$$Y_i = \alpha_0 + \alpha_1 X_i + \alpha_2 D_i + u_i \quad \dots (15.1)$$

where Y_i = 1, if a person owns a house
 = 0, if he/she does not own a house
 X_i = is family income
 D_i = 1, if the person is male, and 0 otherwise.

Equation (15.1) looks like a linear regression model but as the dependent variable (Y_i) is binary, or dichotomous, it is called a linear probability model (LPM). The conditional expectation of Y_i given by X_i and D_i , $E(Y_i|X_i, D_i)$, can be interpreted as the *conditional probability* that the event will occur given X_i and D_i , that is, $\Pr(Y_i = 1|X_i, D_i)$. Thus, in our example, $E(Y_i|X_i, D_i)$ gives the probability of a person owning a house and whose income is given by X_i and whose gender is identified by D_i . Equation (15.1) can be estimated by using the OLS method. The OLS method, however, has certain drawbacks.

The justification of the name LPM for model (15.1) can be seen as follows: By assuming $E(u_i) = 0$ and taking expected value of equation (15.1), we obtain,

$$E(Y_i|X_i, D_i = 1) = \alpha_0 + \alpha_1 X_i + \alpha_2 \quad \dots (15.2)$$

Suppose the probability that the event will occur, that is, $\Pr(Y_i = 1) = p_i$. Then, the probability that the event does not occur $\Pr(Y_i = 0) = (1 - p_i)$.

Taking the expected value

$$\begin{aligned} E(Y_i|X_i, D_i) &= E(\alpha_0 + \alpha_1 X_i + \alpha_2) \\ &= 1(p_i) + 0(1 - p_i) \end{aligned} \quad \dots (15.3)$$

Since the probability p_i must remain between 0 and 1, we have the restriction on the conditional expectation that

$$0 \leq E(Y_i|X_i, D_i) \leq 1 \quad \dots (15.4)$$

The conditional expectation of the model (15.1) can, in fact, be interpreted as the conditional probability of Y_i . In general, the expectation of a Bernoulli random variable is the probability that the random variable equals 1. If there are n independent trials, each with a probability p of success and probability $(1 - p)$ of failure, and X of these trials represent the number of successes, then X is said to follow the binomial distribution. The mean of the binomial distribution is np and its variance is $np(1-p)$. Thus, the OLS method can be easily extended to binary dependent variable regression models. However, the LPM poses several limitations.

15.2.2 Limitations of the Model

The major limitations of the LPM are discussed below.

i) Non-Normality of u_i

Since the dependent variable (X_i) assumes only two values (0 and 1), the error term (u_i) also takes only two values. It implies that the error term follows the Bernoulli distribution. As a result, it is not normally distributed.

Recall from Unit 4 that the OLS method does not require the error term (u_i) to be normally distributed. However, we assumed u_i to be normally distributed for the

purpose of hypothesis testing. But the assumption of normality for u_i is not correct for LPM because, like Y_i , the error term u_i also takes only two values; that is, they also follow the Bernoulli distribution. Non-fulfillment of the normality assumption may not be so critical as it appears, because for obtaining unbiased OLS estimators, u_i need not be normally distributed. Besides, as the sample size increases indefinitely, statistical theory shows that the OLS estimators tend to be normally distributed. As a result, in large samples the statistical inference of the LPM will follow the usual OLS procedure under the normality assumption.

ii) Heteroscedasticity of u_i

Remember that for a Bernoulli distribution the mean and variance are p_i and $p_i(1 - p_i)$ respectively. Here, p_i is the probability of success, which implies that the variance is a function of the mean. Thus, the error variance is heteroscedastic. However,

$$P_i = E(Y_i | X_i, D_i = 1) = \alpha_0 + \alpha_1 X_i + \alpha_2 \quad \dots (15.5)$$

In (15.5), the mean of Y_i , that is, P_i depends on the explanatory variable X_i . Therefore, $var(u_i)$ also depends on the explanatory variables.

iii) Value of $\hat{Y}_i$ remains outside the range

We mentioned earlier that $\hat{Y}_i$ indicates the probability of occurrence, p_i . However, the OLS estimator gives no guarantee that the probability will be between 0 and 1 (See Fig. 15.1). This is because the probability increases linearly with explanatory variables.

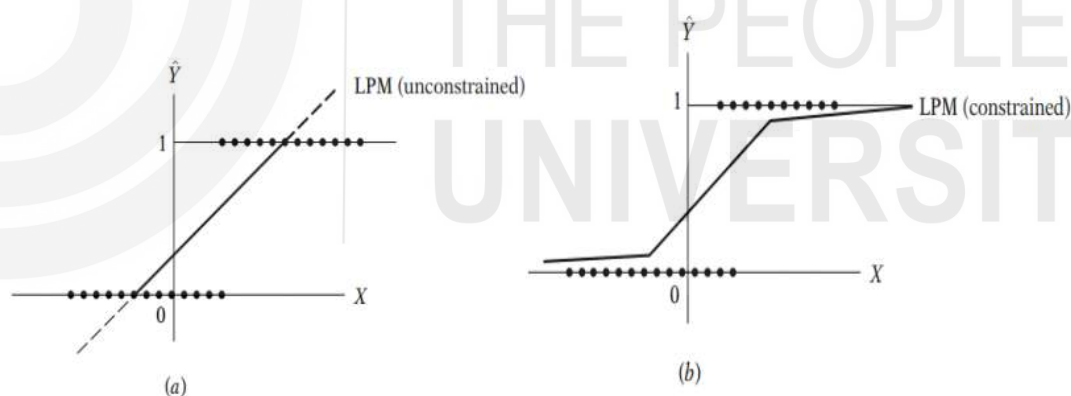


Fig. 15.1: Linear Probability Models

There are two ways to get rid of this problem: (i) Estimate the LPM by the usual OLS method and find out whether the estimated $\hat{Y}_i$ remains between 0 and 1. If for some observations $\hat{Y}_i$ is less than 0 (that is, negative), we assume them to be zero. Similarly, for some observations, if $\hat{Y}_i$ is greater than 1, we assume them to be 1. The second solution is to adopt some estimation technique such as Logit and Probit, that will guarantee that the estimated conditional probabilities $\hat{Y}_i$ remains between 0 and 1.

The scatter plot depicted in Fig. 15.1 is drawn on the basis of data given in Table 15.1. We observe that for some values of X_i , the predicted probability ($\hat{P}_i$) lies

outside the limiting values, i.e., $0 \leq \hat{P}_i \leq 1$. This violates an important axiom of probability theory that probability ranges between 0 and 1.

Table 15.1: Hypothetical Data on House Ownership Status

Household	Y	X	Household	Y	X
1	0	8	21	1	24
2	1	20	22	0	16
3	1	18	23	0	12
4	0	11	24	0	11
5	0	12	25	0	16
6	1	19	26	0	11
7	1	20	27	1	20
8	0	13	28	0	18
9	1	20	29	0	11
10	0	10	30	0	10
11	1	17	31	1	17
12	1	18	32	0	13
13	0	14	33	1	21
14	0	25	34	1	20
15	1	6	35	0	11
16	1	19	36	0	8
17	1	16	37	1	17
18	0	10	38	1	16
19	0	8	39	0	7
20	1	18	40	1	17

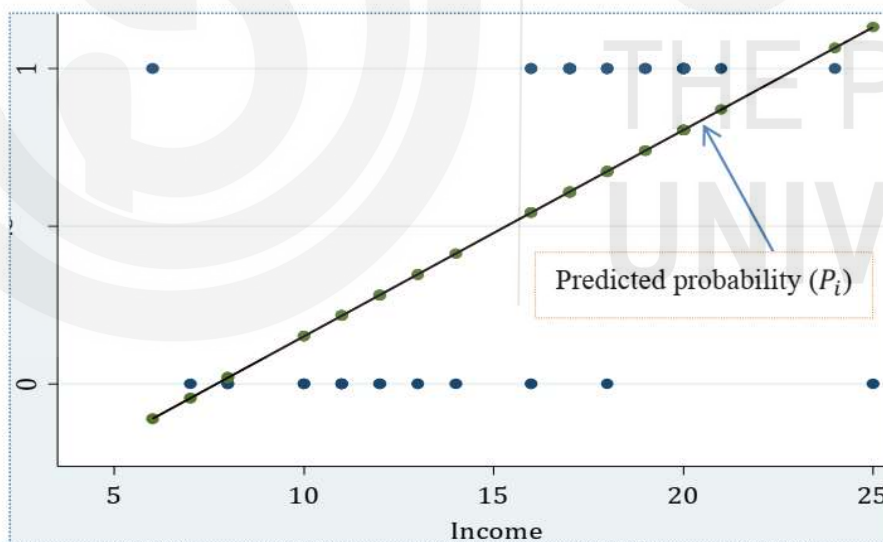


Fig. 15.2: Scatter Plot of Y_i and $\hat{Y}_i$ in LPM

iv) R^2 is not Appropriate

Corresponding to the value of explanatory variables (X), the dependent variable (Y) is either 0 or 1. Therefore, all the Y values will either lie along the X - axis or along the line corresponding to Y equals 1. Therefore, LPM is not expected to fit such a scatter plot very well. As a result, computed R^2 is of limited value in the dichotomous response models.

In Fig. 15.1, we plotted the actual values Y_i , and the predicted values $\hat{Y}_i$. Corresponding to a given X , the value of Y is either 0 or 1. Therefore, all the Y values will either lie along the X axis or along the line corresponding to 1. That is, when we regress Y_i on the given explanatory variable (X_i) in the case of linear regression model, we will get the fitted regression line given above (i.e., Fig. 15.1). Therefore, generally LPM is not expected to fit such a scatter diagram well. As a result, the conventionally computed R^2 is likely to be much lower for such models. In most practical applications, the R^2 ranges between 0.2 to 0.6.

15.2.3 Interpretation of LPM

(Example 15.1)

To illustrate the method of Linear Probability Model and the points raised above, let us assume that we want to study house ownership status (Y) of households in a given town. Assume that we have data for 40 households on house ownership status and income of households (X) (in thousands of rupees) (see Table 15.1).

Based on the data given in Table 15.1, we estimate the LPM by the OLS method. The results obtained are as follows:

$$\begin{aligned}\hat{Y} &= -0.5 + 0.0652 X_i & \dots (15.6) \\ \text{SE} &= (0.2096) \quad (0.0134) \\ t\text{-ratio} & (-)2.39 \quad 4.88 & R^2 = 0.3854\end{aligned}$$

Let us interpret the above result.

The intercept of $(-) 0.5$ gives the probability that a household with zero income will own a house. Since this value is negative, and since probability cannot be negative, we treat this value as zero. The slope value of 0.0652 means that for a unit change in income (here, Rupees 1000), the probability of owning a house increases by 0.0652 or 6.52 percent.

Table 15.2: Actual Y_i and predicted $\hat{Y}_i$

Y	$\hat{Y}_i$	Y	$\hat{Y}_i$	Y	$\hat{Y}_i$	Y	$\hat{Y}_i$
0	0.022	1	0.609	1	1.065	1	0.609
1	0.804	1	0.674	0	0.543	0	0.348
1	0.674	0	0.413	0	0.280	1	0.869
0	0.217	0	1.130	0	0.217	1	0.804
0	0.283	1	-0.109	0	0.543	0	0.217
1	0.739	1	0.739	0	0.217	0	0.022
1	0.804	1	0.543	1	0.804	1	0.609
0	0.348	0	0.152	0	0.674	1	0.543
1	0.804	0	0.022	0	0.217	0	-0.043
0	0.152	1	0.674	0	0.152	1	0.609

Table 15.2 shows the estimated probabilities, $\hat{Y}_i$, for various income levels. The most noticeable feature of Table 15.2 is that two estimated values are negative,

and two values are greater than 1. This is one reason why the LPM is not recommended.

Check Your Progress 1

- 1) A linear probability model of a hypothetical data set (sample size $n = 200$) yields the following results:

$$\hat{Y}_i = \frac{-0.084}{(0.021)} + \frac{0.0017}{(0.0004)} \text{Income}_i + \frac{0.008}{(0.0032)} D_i$$

$$t = (-4.00) \quad (4.25) \quad (2.5) \quad R^2 = 0.882$$

$Y_i = 1$ for ownership of a house, and 0 otherwise; $D_i = 1$ for male and 0 for female; income is in Rs.'000 and values in parentheses are standard errors and t-ratios.

- Interpret the model.
- Interpret the intercept term.
- Interpret the coefficient of the variable 'income'.
- Interpret the coefficient of the dummy variable, D_i .

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- 2) What are the limitations of the linear probability model?

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15.3 LOGIT MODEL

As we saw in the previous section, predicted values of the LPM model could be outside the range. Therefore, instead of depending on the LPM, we need a (probability) model that has the following two features:

- As X_i increases, $p_i = E(Y_i = 1|X)$ increases but never steps outside the $[0 - 1]$ interval.
- The relationship between p_i and X_i is non-linear. As X_i gets smaller, $\hat{Y}_i$ tends to 0 (that is, it approaches 0 at slower and slower rate). On the other hand, as X_i gets very large, $\hat{Y}_i$ tends to 1.

The logit model (logit is an abbreviation for logistic unit) is an option before us. In this model, we can limit the estimated probabilities inside the (0–1) range. In the logit model (as well as the probit model, which we will study in the next

section), we assume that there is an underlying latent variable (y^*) which is not observable, but it manifests in a dichotomous realization. For example, the latent variable is ‘the ability to get employed’, which is not observable. What we observe is whether a person is employed ($y = 1$) or not employed ($y = 0$). Thus, $y_i = 1, \text{ if } y^* > 0$ and $y_i = 0, \text{ if } y^* \leq 0$.

In the logit model, the probability of occurrence of the event is defined as

$$p_i = E(Y = 1|X_i) = \frac{1}{1+e^{-(\alpha_0+\alpha_1X_i)}} \quad \dots (15.7)$$

If we let $Z_i = \alpha_0 + \alpha_1X_i$, then equation (15.7) can be expressed as

$$p_i = E(Y = 1|X_i) = \frac{1}{1+e^{-Z_i}} = \frac{e^{Z_i}}{1+e^{Z_i}} \quad \dots (15.8)$$

Equation (15.8) is known as the (cumulative) logistic distribution function which satisfies the two conditions stated in the beginning of this Section. Since p_i gives the probability of owning a house, the probability of ‘not owning a house’, $(1 - p_i)$, is:

$$1 - p_i = \frac{1}{1+e^{Z_i}} \quad \dots (15.9)$$

If we divide equation (15.8) by equation (15.9), we obtain

$$\frac{p_i}{1-p_i} = \frac{e^{Z_i}/(1+e^{Z_i})}{1/(1+e^{Z_i})} = e^{Z_i} = e^{\alpha_0+\alpha_1X_i} \quad \dots (15.10)$$

Equation (15.10) is the ‘Odds-Ratio’ in favour of owning a house. It is the ratio of the probability that ‘a family will own a house’ to the probability that ‘it will not own a house’. Thus, for example, if $p_i = 0.8$, the odds-ratio is 4. It implies that the odds are 4 to 1 (since $\frac{p_i}{1-p_i} = \frac{0.8}{1-0.8} = \frac{0.8}{0.2} = 4$) in favour of the family owning a house.

If we take the natural logarithm of equation (15.10), we obtain

$$L_i = \ln\left(\frac{p_i}{1-p_i}\right) = \ln\left(\frac{1+e^{Z_i}}{1+e^{-Z_i}}\right) = Z_i = \alpha_0 + \alpha_1X_i \quad \dots (15.11)$$

The ‘log of the odds-ratio’ (L_i), given in equation (15.11) is linear in parameters and linear in variables.

15.3.1 Features of Logit Model

We point out the features of the logit model below.

- (i) In equation (15.11), as Z varies from $-\infty$ to $+\infty$, the ‘log of the odds-ratio’ (L_i), varies from $-\infty$ to $+\infty$. That is, although the probability remains between 0 and 1, the explanatory variable can take any value between $-\infty$ and $+\infty$.
- (ii) Although L is linear in X , the probability (p_i) is not linear. Although we have included only a single explanatory variable, in the preceding model, we can add as many explanatory variables as required by the underlying theory.

- (iii) The interpretation of the logit model given in equation (15.11) is as follows: The slope coefficient measures the change in L for a unit change in the explanatory variable. The slope coefficient tells how the log-odds in favor of owning a house change as income changes by a unit. The intercept is the value of the log-odds in favor of owning a house if income is zero. Like most interpretations of intercepts, this interpretation may not have any physical meaning.
- (iv) Remember that the odds-ratio is not the probability of the occurrence of the event (such as owning a house). Suppose we want to find out the probability of owning a house at a certain level of income, say, X^* , from equation (15.7). The question is: How do we estimate α_0 and α_1 . The answer is given below.

15.3.2 Estimation of Logit Model

Let us add a stochastic error term to equation (15.11). We obtain

$$L_i = \ln\left(\frac{p_i}{1-p_i}\right) = \alpha_0 + \alpha_1 X_i + u_i \quad \dots (15.12)$$

To estimate equation (15.12), we need data of X_i and logit (L_i). This depends on the type of data we have for analysis. There could be two types of data:

- (i) Data at the individual or micro level and
- (ii) Grouped or replicated data.

(i) Data at the individual or micro level

From equation (15.12), $p_i = 1$ if a family owns a house, and $p_i = 0$ if it does not own a house. Thus,

$$L_i = \ln\left(\frac{1}{1-1}\right) = \ln\left(\frac{1}{0}\right), \text{ if a person owns a house}$$

$$L_i = \ln\left(\frac{0}{1-0}\right) = \ln\left(\frac{0}{1}\right), \text{ if a person does not own a house}$$

These imply that if we have data at the micro or, individual level, we cannot estimate equation (15.12) by the standard OLS (ordinary least squares). In this situation, we may have to resort to **non-linear estimation procedure** using the maximum likelihood (ML) method.

(ii) Grouped or Replicated Data

Here, for a given explanatory variable, observations with equal values are grouped together. Such a model is called Grouped Logit or GLogit. For example, in the model represented by equation (15.12), people who earn the same disposable income are grouped together. Corresponding to each group of income level X_i , there are N_i families, among whom n_i are homeowners ($n_i < N_i$). Therefore,

$$\hat{p}_i = \frac{n_i}{N_i} \quad \dots (15.13)$$

gives the *relative frequency* house owners and can be used as an estimate of the true P_i corresponding to each X_i . This true P_i is used to estimate the logit, $\hat{L}_t$.

$$\text{Thus, } \hat{L}_i = \ln\left(\frac{\hat{p}_i}{1-\hat{p}_i}\right) = \hat{\alpha}_0 + \hat{\alpha}_1 X_i \quad \dots (15.14)$$

gives a fairly good estimate of the true logit, L_i , if the number of observations N_i at each X_i is reasonably large.

Notes on the estimation of Logit model:

- Logit models are estimated using maximum likelihood method (MLM).
- Standard errors (SE) are asymptotic (i.e., applicable for large samples). Hence, we have to use Z statistic instead of t-statistic.
- R-squared is not meaningful in logit models.
- Likelihood ratio (LR) test is applicable.

15.3.2 Interpretation of Logit Model

(Example 15.2)

Table 15.3: Cross-sectional Data on GPA, ASE, PC and GRADE

Students	GPA	ASE	PC	GRADE (A=1, rest=0)
1	2.66	20	0	0
2	2.89	22	0	0
3	3.28	24	0	0
4	2.92	12	0	0
5	4	21	0	1
6	2.86	17	0	0
7	2.76	17	0	0
8	2.87	21	0	0
9	3.03	25	0	0
10	3.92	29	0	1
11	2.63	20	0	0
12	3.32	23	0	0
13	3.57	23	0	0
14	3.26	25	0	1
15	3.53	26	0	0
16	2.74	19	0	0
17	2.75	25	0	0
18	2.83	19	0	0
19	3.12	23	1	0
20	3.16	25	1	1
21	2.06	22	1	0
22	3.62	28	1	1
23	2.89	14	1	0
24	3.51	26	1	0
25	3.54	24	1	1
26	2.83	27	1	1
27	3.39	17	1	1
28	2.67	24	1	0
29	3.65	21	1	1
30	4	23	1	1
31	3.1	21	1	0
32	2.39	19	1	1

Suppose that a Mathematical Economics instructor gave routine weekly exercises to her section “A” students in the previous semester. She wants to study the effect of the performance in these routine exercises on the final grade of students in the

course, Mathematical Economics. At the end of the semester, she found average scores in exercise (ASE) for each student.

The dependent variable is the final grade of students for Mathematical Economics ($Y_i = 1$, if the student scores A and $Y_i = 0$, otherwise). Other explanatory variables include the Grade Point Average (GPA) of students and Personal –Computer ownership (PC = 1, if the student has a PC and, PC = 0, if the student does not have a PC).

Estimation of the Logit model gives the following results:

$$L_i = \alpha_0 + \alpha_1 \text{GP} + \alpha_2 \text{ASE} + \alpha_3 \text{PC} + u_i$$

We present the estimation procedure and interpretation of the model defined above. We can interpret the regression results using: **Logit, Odds-ratio and marginal effect.**

A. Logit Interpretation of the Estimated Logit Model

Table 15.4 below presents the logit regression result obtained using econometric software Stata14.

Table 15.4: Logistic Regression Result

Logistic regression				Number of obs	=	32
Log likelihood = -12.889633				LR chi2(3)	=	15.40
				Prob > chi2	=	0.0015
				Pseudo R2	=	0.3740
grade	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	2.826113	1.262941	2.24	0.025	.3507938	5.301432
ase	.0951577	.1415542	0.67	0.501	-.1822835	.3725988
pc	2.378688	1.064564	2.23	0.025	.29218	4.465195
_cons	-13.02135	4.931325	-2.64	0.008	-22.68657	-3.35613

We interpret the coefficients of the independent variables of the logit regression results as log of the odds-ratio. Coefficients of GPA and ASE in the Table above measure the change in the estimated logit for a unit change in the value of explanatory variable (holding other explanatory variables constant). Thus,

- ✓ The GPA coefficient of 2.826 means, with other variables held constant, that if GPA increases by a unit, on an average, the estimated logit increases by about 2.826 units, suggesting a positive relationship between the two. Holding other factors constant, as GPA increases by one point, the average logit value goes up by about 2.83, that is, the log of odds in favor of scoring “A” grade increases by about 2.83 and it is statistically significant as implied by the P-value of the Z-test.

- ✓ Holding other factors constant, as ASE increases by one point, the log of the odds-ratio increases by 0.095, but this effect is statistically not significant.
- ✓ Among students, students who have a PC compared to those who do not have PC, the log of odds in favor of scoring “A” grade increases by about 2.38 *and* it is statistically significant as implied by the P-value of the Z-test.

As you can see, all the explanatory variables have a positive effect on the logit, although statistically the effect of ASE is not significant. However, together all the explanatory variables have a significant effect on scoring “A” grade, as the LR statistic (an equivalent to F-test in linear regression) is 15.40, whose ***P-value*** is about 0.0015, which is less than 5 per cent.

Note, however, that a more meaningful interpretation is in terms of the odds-ratio, which are obtained by taking the antilog of the various coefficients of the logit model. Thus, if you take the antilog of the GPA coefficient of 2.826 you will get 16.8778 ($= \exp^{(2.826)}$). As GPA of a student rises by a unit, then the student is about 16.88 times likely to get Grade “A”, other things remaining the same. Using Stata, we can easily estimate the odds ratio for all explanatory variables as given in Table 15.5.

B. Odds-Ratio Interpretation

Table 15.5 below presents the odds-ratio regression result of the logit model.

Table 15.5: Odds Ratio Regression Result

Logistic regression				Number of obs	=	32
				LR chi2(3)	=	15.40
				Prob > chi2	=	0.0015
Log likelihood = -12.889633				Pseudo R2	=	0.3740
grade	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	16.87972	21.31809	2.24	0.025	1.420194	200.6239
ase	1.099832	.1556859	0.67	0.501	.8333651	1.451502
pc	10.79073	11.48743	2.23	0.025	1.339344	86.93802
_cons	2.21e-06	.0000109	-2.64	0.008	1.40e-10	.03487

The odds-ratio interpretation is as follows:

- Holding other variables constant, as GPA increases by one point, the probability of scoring Grade “A” is 16.87 times the probability of getting other grades (B, C, D, Fx, F).
- Holding other variables constant, as ASE increases by one point, the probability of scoring Grade “A” is about 1.1 times the probability of getting other grades (B, C, D, Fx, F) but it is statistically not significant.

- Holding other variables constant, students who have PC are 10.79 times more likely to score Grade “A” than other grades compared to non-PC owners, and it is statistically significant at 5% level of significance.

C. Marginal Effect Interpretation

Since the language of logit and odds ratio may be unfamiliar to some, we always compute the probability of something happening. Table 15.6 below presents the probability / marginal effect regression result of the logit model.

Table 15.6: Marginal Effects of Logit Regression Result

Marginal effects after logit							
y = Pr(grade) (predict)							
= .25282025							
variable	dy/dx	Std. Err.	z	P> z	[	95% C.I.	]
gpa	.5338589	.23704	2.25	0.024	.069273	.998445	3.11719
ase	.0179755	.02624	0.69	0.493	-.033448	.069399	21.9375
pc*	.4564984	.18105	2.52	0.012	.10164	.811357	.4375
(*) dy/dx is for discrete change of dummy variable from 0 to 1							

The marginal effect /probability interpretation of the estimated logit model given above is as follows:

- As GPA increases by one point, the probability of scoring grade “A” by an average student increase by 53%, if the other factors score average.
- As ASE increases by one point, the probability of scoring grade “A” by an average student increase by about 1.8%, assuming that the other factors score average.
- For a student who owns PC, the probability of scoring grade “A” is higher by 45.6% as compared to non-PC owners, assuming that the other factors score average.

Check Your Progress 2

1) Derive the odds-ratio for the logit model.

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2) How do you interpret the results of the logit model?

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3) Explain how the marginal effects of the logit model is interpreted.

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15.4 PROBIT MODEL

The probit model is usually used if the cumulative density function (CDF) is normal. On the other hand, the logit model uses the cumulative logistic function. In principle we can substitute the normal CDF in place of the logistic CDF.

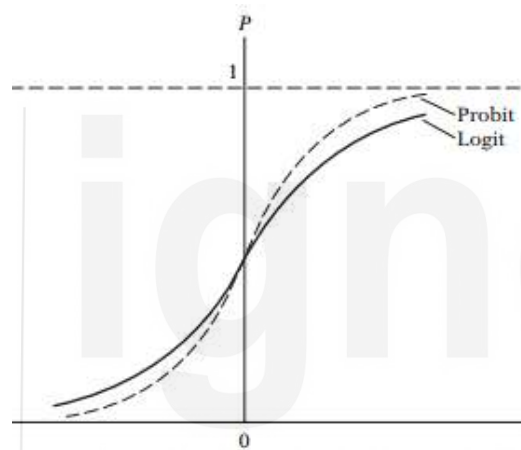


Fig. 15.3: Comparison between Logit and Probit Cumulative Distributions

Fig. 15.3 portrays the difference between the cumulative distribution function (CDF) of Logit and Probit models. As the value of the explanatory variable (X) increases, the probability of occurrence of the dependent variable increases faster (gets closer to 1) under probit model than under logit model. Similarly, as the value of X gets smaller and smaller, the probability of the dependent variable not-to-occur decreases faster (gets closer to 0) under probit than under logit. Beyond this, they are quite similar. Yet, in principle, the normal CDF can be substituted in place of the logistic CDF because of this pretty resemblance between them.

15.4.1 Specification of Probit Model

Suppose the decision of ownership of a house by a person or family depends on the utility obtained from owning the house. Mathematically,

$$I_i = \alpha_0 + \alpha_1 X_i \quad \dots (15.16)$$

where, " I_i " is *unobservable* utility index, also called **latent variable**, whose higher values are associated with higher probability of owning a house. Here, X_i is the income of i^{th} person or family. Now, assume that there is a threshold or critical level, say I_i^* , which itself is unobservable, such that if I_i **exceeds** I_i^* the family will **own a house**, otherwise it will not. If we assume further that I_i^* , like

I_i , is *normally* distributed with the same mean and variance, it is possible not only to estimate the parameters of the index given in equation (15.16) but also to get some information about the unobservable index itself.

The probability that I_i^* is **less** than or equal to I_i can be computed from the standardized normal CDF as:

$$P_i = P(Y = 1 | X) = P(I_i^* \leq I_i) = P(Z_i \leq \alpha_0 + \alpha_1 X_i) = F(\alpha_0 + \alpha_1 X_i) \quad \dots (15.17)$$

where, $P(Y = 1 | X)$ means the probability that an event occurs given the value(s) of the explanatory variables, Z_i is the standard normal variable, i.e., $Z_i \sim N(0, \sigma^2)$. Note that F is the standard normal CDF, which is:

$$\begin{aligned} F(I_i) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{I_i} e^{-\frac{z^2}{2}} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\alpha_0 + \alpha_1 X_i} e^{-\frac{z^2}{2}} dz \quad \dots (15.18) \end{aligned}$$

The **area between** $-\infty$ and I_i measures the probability of **owning a house**. If we take the inverse of equation (15.18), we get information on I_i, α_0, α_1 . That is,

$$I_i = F^{-1}(I_i) = F^{-1}(P_i) = \alpha_0 + \alpha_1 X_i \quad \dots (15.19)$$

Where, F^{-1} is the inverse of the normal CDF.

In general, the probit model can be specified as follows:

$$Pr(Y_i = 1 | X_i) = Pr(I_i \geq I_i^*) = F(Z) = P_i = \int_{-\infty}^Z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2)} dz \quad \dots (15.20)$$

This model is called the probit model and can be estimated using maximum likelihood method (MLM). The coefficients from the probit model are difficult to interpret because they measure the change in the unobservable I_i associated with a change in one of the explanatory variables. A more useful measure is the marginal effects.

15.4.2 Estimation of Probit Model

Estimation of parameters and I_i depends on types of data we deal, meaning that whether we have grouped data or ungrouped data set.

(a) Estimation with Grouped Data (gprobit)

In the case of the logit model, we get the relative frequency, (the empirical measure of probability) of owning a house at various income level, but here in probit model, I_i can be obtained from normal CDF. Obtaining I_i makes estimating parameters relatively straightforward.

(b) Estimation with Ungrouped or Individual Data

The problem we face in Logit regression is also presents here. For this reason, we will have to use a non-linear estimating procedure based on the *method of maximum likelihood*.

15.4.3 Interpretation of Probit Model

(Example 15.3)

Let us revisit our Grade example data given in Table 15.3. It gives data on 32 students about their final grade in Mathematical Economics examination in relation to the variables GPA, ASE, and PC. Let us see what the Probit results look like. The regression results based on the method of maximum likelihood using Stata14 are given in Table 15.7.

Table 15.7: Probit Regression Result

Probit regression						Number of obs	=	32
						LR chi2(3)	=	15.55
						Prob > chi2	=	0.0014
Log likelihood = -12.818803						Pseudo R2	=	0.3775
grade	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]			
gpa	1.62581	.6938825	2.34	0.019	.2658255	2.985795		
ase	.0517289	.0838903	0.62	0.537	-.1126929	.2161508		
pc	1.426332	.5950379	2.40	0.017	.2600795	2.592585		
_cons	-7.45232	2.542472	-2.93	0.003	-12.43547	-2.469166		

Estimated coefficients do not quantify the influence of the explanatory variables on the probability that the dependent variable takes on the value one because these coefficients are parameters of the latent model. Thus, there is no standard and convincing way to interpret the coefficients from the output of a probit regression. We need to interpret the *marginal effects* of the explanatory variables, that is, how much the (conditional) probability of the outcome variable changes when you change the value of an explanatory variable, holding all other explanatory variables constant at some values. As far as the coefficients of a Probit model are concerned, what we need to do is to explain the implication of the coefficients. For instance,

- The coefficient of GPA is significantly different from zero (at 5% level) indicating relevant relationship between scoring “A” grade and student’s GPA.
- The coefficient of PC is significantly different from zero (at 5% level) indicating relevant relationship between scoring “A” grade and PC ownership.

In short, students with higher GPA and PC are more likely to score “A” compared to their counterparts. In general, “qualitatively”, the results of the Probit model are comparable with those obtained from the logit model in that GPA and PC are individually statistically significant while the variable ASE is

statistically not significant in both models. Collectively, all the coefficients are statistically significant, since the value of the **LR** statistic is 15.5458 with a p value of 0.0014.

Table 15.8: Marginal Effect of the Probit Model

Marginal effects after probit							
y = Pr(grade) (predict)							
= .26580809							
variable	dy/dx	Std. Err.	z	P> z	[	95% C.I.	]
gpa	.5333471	.23246	2.29	0.022	.077726	.988968	3.11719
ase	.0169697	.02712	0.63	0.531	-.036184	.070123	21.9375
pc*	.464426	.17028	2.73	0.006	.130682	.79817	.4375
(*) dy/dx is for discrete change of dummy variable from 0 to 1							

The marginal effect results and their interpretation in a Probit model are similar to that of logit model.

Therefore, the **marginal effect / probability** interpretation of the estimated Probit model presented above is as follows:

- As GPA increases by one unit, the probability of scoring grade “A” by an average student increases by 53%, holding other factors constant.
- As ASE increases by one point, the probability of scoring grade “A” by an average student increases by about 1.7%, holding other factors constant.
- For a student who owns PC the probability of scoring grade “A” is higher by 45.6% as compared to non-PC owners, holding other factors constant.

15.5 JOINT SIGNIFICANCE IN QUALITATIVE RESPONSE REGRESSION MODELS

As far as joint significance is concerned, as equivalent of the **F** test (in the linear regression model) there are various ways of testing multiple restrictions in Probit and Logit models.

The most commonly used test, and most easily calculated test is the **Likelihood Ratio (LR) Test**. It is used when we wish to test our exclusion restrictions, i.e. whether we should or should not exclude a set of variables. The idea is a simple one: we are maximizing is the log likelihood function, and since variables are excluded from the regression relationship, the objective function declines. The question then is whether we have a significant fall in the log likelihood function value. Like the F-test in linear regression models, in LR test the joint hypothesis to be tested is that all the explanatory variables are simultaneously irrelevant versus at least one of the explanatory variables is relevant. The likelihood ratio statistic is given as follows:

$$LR = 2 (L_{UR} - L_R) \quad \dots (15.21)$$

where, L_{UR} and L_R are the log likelihoods for the unrestricted and restricted models.

Given the null hypothesis, the LR statistic follows the chi-square (χ^2) distribution asymptotically with degrees of freedom equal to the number of explanatory variables.

15.6 GOODNESS-OF-FIT IN LOGIT AND PROBIT MODELS

A goodness-of-fit measure is a summary statistic indicating the accuracy with which the model approximates the observed data. However, the conventional measure of goodness of fit, R^2 , is not particularly meaningful in binary dependent variable models. In the case in which the dependent variable is qualitative, accuracy can be judged either in terms of the fit between the calculated probabilities and observed response frequencies or in terms of the model's ability to forecast observed responses. Measures similar to R^2 , called **pseudo- R^2** , are available. Note, however, that contrary to the linear regression model, there is no single measure for the goodness-of-fit in binary choice models and a variety of measures exists. The most common goodness of fit measure is the one that is proposed by **McFadden** (1974), which is defined as:

$$\text{Pseudo-}R^2 = \{1 - (L_{UR}/L_R)\} \quad \dots (15.22)$$

where, as before L_{UR} is the log likelihood value you obtain from the unrestricted model, and L_R is that generated by a regression (*either Probit or Logit. But note that this goodness of fit measure is also used and reported often by statistical packages for any regression that involves MLE*) with only the intercept.

- The Pseudo- R^2 measures the fit using the likelihood function and measures the improvement in the value of the log likelihood relative to having no explanatory variables (X_i). For instance, in our estimated Logit and Probit models above, we see that the Pseudo- $R^2 = 0.37$. This suggests that the log-likelihood value increases by about 37% with the introduction of the set of explanatory variables in the models.
- Similar to the R^2 of the linear regression model, it holds that $0 \leq \text{Pseudo-}R^2 \leq 1$. An increasing Pseudo- R^2 may indicate a better fit of the model, whereas there is no simple interpretation, as in the case of the classical regression model.

Check Your Progress 3

- 1) Bring out the salient features of the probit model.

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2) How do you interpret the pseudo-R-squared?

Pressure Groups

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15.7 LET US SUM UP

Here is a summary of linear probability model (LPM), logit model, and probit model:

Linear probability model (LPM): The LPM is a simple linear regression model that can be used to predict a binary outcome. The dependent variable is a dichotomous variable, such as "yes" or "no, " "success" or "failure, " or "purchased" or "not purchased." The independent variables can be continuous or categorical. The LPM assumes that the error term follows a normal distribution.

- The LPM is a simple linear regression model where the dependent variable is a binary variable.
- The coefficients in the LPM can be interpreted as the change in the mean of the dependent variable for a one-unit change in the independent variable.
- However, the LPM does not consider the fact that the dependent variable is binary, and this can lead to biased estimates of the coefficients.

Logit model: The logit model is a more sophisticated model than the LPM. It assumes that the dependent variable follows a logistic distribution. The logistic distribution is a skewed distribution that is centered at 0.5. The logit model can provide more accurate predictions than the LPM, especially when the dependent variable is close to 0 or 1.

- The logit model is a nonlinear regression model where the dependent variable is a binary variable.
- The coefficients in the logit model can be interpreted as the change in the log-odds of the dependent variable for a one-unit change in the independent variable.
- The logit model is more appropriate for modeling binary variables than the LPM because it considers the fact that the dependent variable can only take on two values.

Probit model: The probit model is another type of model that can be used to predict a binary outcome. The probit model assumes that the dependent variable follows a normal distribution. The probit model is like the logit model in terms of its predictive accuracy. However, the probit model is sometimes preferred because it is easier to interpret than the logit model.

- The probit model is another nonlinear regression model where the dependent variable is a binary variable.
- The coefficients in the probit model can be interpreted as the change in the probability of the dependent variable for a one-unit change in the independent variable.
- The probit model is like the logit model, but it assumes that the error term follows a normal distribution instead of a logistic distribution.

15.8 KEY WORDS

Binary Model	:	A binary model is a statistical model that predicts a categorical dependent variable with two possible outcomes. The dependent variable is typically coded as a binary variable, such as 0 or 1, or 'yes' or 'no'. The independent variables can be continuous or categorical.
Qualitative Dependent Variable	:	A qualitative dependent variable is a variable that can take on one of a limited number of categories, such as yes or no, high or low, or agree or disagree. It is a type of categorical variable. The qualitative variable is not continuous.
Dichotomous Variables	:	A dichotomous variable is a variable that can only take on two values, such as "yes" or "no," "true" or "false," or "male" or "female." Dichotomous variables are often used in surveys and experiments, where the goal is to understand the relationship between two or more variables.
Odds Ratio	:	It is calculated by dividing the odds of occurrence of an event by the odds of the event occurring in another group. Odds ratios are always positive. Odds ratios can be greater than 1 or less than 1. An odds ratio of 1 indicates no association between the two variables. An odds ratio greater than 1 indicates that the event is more likely to occur in the first group than in the second group. An odds ratio less than 1 indicates that the event is less likely to occur in the first group than in the second group.
Marginal Effect	:	Marginal effects are calculated at a <i>specific point</i> in the data. The point at which the marginal effect is calculated is called the base point. Marginal effects can be <i>positive or negative</i> . A positive marginal effect indicates that an increase in the independent variable is associated with an increase in the dependent variable. A negative marginal effect indicates that an increase in the independent variable is associated with a decrease in the dependent variable. Marginal effects can be interpreted in different ways depending on the <i>context</i> . For example, a marginal effect of

0.10 in a logistic regression model may be interpreted as a 10% increase in the probability of the event occurring.

15.9 SOME USEFUL BOOKS

- Baltagi, Badi H. (2008), *Econometrics*, Fourth Edition, Springer-Verlag Berlin Heidelberg.
- Berndt, Ernst R. (1991), *The Practice of Econometrics: Classic and Contemporary*, Addison-Wesley.
- Cameron, Colin and Pravin Trivedi (2005), *Microeconometrics*, Cambridge: Cambridge University.
- Maddala, G. S. (1983), *Limited Dependent and Qualitative Variables in Econometrics*, Cambridge University Press, Cambridge.
- Verbeek, Marno (2017), 'A Guide to Modern Econometrics,' Fifth Edition, John Wiley and Sons Ltd.
- Wooldridge, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

15.10 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) (a) Since all t-values (coefficient divided by standard error) are greater than 2, all the coefficients are statistically significant at 5 per cent level of significance.
- (b) The intercept term gives the “probability” that a female person (since $D_i = 0$ for female) with zero income will own a house. This value is negative. Since probability cannot be negative, we may treat this value as zero.
- (c) The slope value of 0.0017 means that for a unit change in income, on average, the probability of owning a house increases by 0.0017 or about 0.17 percent. Of course, given a particular level of income, we can estimate the actual probability of owning a house. If, for example, income = Rs. 500, the estimated probability of owning a house is
- $$\begin{aligned}(\hat{Y}_i | X_i = 500, D_i = 1) &= -0.084 + 0.0017 * 500 + 0.008 * 1 \\ &= 0.846 = 84.6 \text{ percent.}\end{aligned}$$
- $$\begin{aligned}(\hat{Y}_i | X_i = 500, D_i = 0) &= -0.084 + 0.0017 * 500 + 0.008 * 0 \\ &= 0.766 = 76.6 \text{ percent.}\end{aligned}$$
- (d) The coefficient of variable D_i is 0.008. It implies that for a male (when $D_i = 1$), the intercept increases by 0.008. Therefore, the probability of owning a house for males is greater than that for females by 0.8 percent.

- 2) Go through Section 15.2.2. You have to summarise the points given there and answer.

Check Your Progress 2

- 1) The log of odds-ratio is given by $L_i = \ln\left(\frac{p_i}{1-p_i}\right) = \alpha_0 + \alpha_1 X_i + u_i$.
- 2) You can interpret the results of the logit model through its Logit, Odds-ratio and marginal effect. See Sub-Section 15.3.2.
- 3) Go through Sub-Section 15.3.2 and answer.

Check Your Progress 3

- 1) Go through Section 15.4 and answer.
- 2) Go through Section 15.6 and answer.



UNIT 16 INTRODUCTION TO SIMULTANEOUS EQUATIONS MODELS*

Structure

- 16.0 Objectives
- 16.1 Introduction
- 16.2 Some Examples of Simultaneous Equations Models
- 16.3 Endogenous Variables and Exogenous Variables
- 16.4 Simultaneity Bias
 - 16.4.1 Endogeneity Problem
 - 16.4.2 The OLS Estimator is Biased
 - 16.4.3 The OLS Estimator is not Consistent
 - 16.4.4 Solution to Simultaneity Bias
- 16.5 Structural Form and Reduced Form
- 16.6 Concept of Identification
 - 16.6.1 Paradox of Identification
 - 16.6.2 Identification Status of an Equation
- 16.7 Identification Conditions
 - 16.7.1 Order Condition of Identification
 - 16.7.1 Rank Condition of Identification
- 16.8 Let Us Sum Up
- 16.9 Key Words
- 16.10 Some Useful Books/References
- 16.11 Answers/Hints to Check Your Progress Exercises
- 16.12 Practice Questions

16.0 OBJECTIVES

After going this Unit, you will be able to

- analyse the nature of simultaneous relationship between variables;
- distinguish between endogenous variables and pre-determined variables;
- explain the concept of simultaneity bias and its implications;
- examine the identification status of an equation in a simultaneous equations system; and
- apply the order and rank conditions for identification of an equation.

*Adapted from Units 13 and 14 of IGNOU SLM, MECE 001: Econometric Methods, written by Prof. Biswajit Nag and Ms. Satavisha Mukherjee, Indian Institute of Foreign Trade (IIFT), New Delhi

16.1 INTRODUCTION

The econometric models we studied so far talk about single equation regression models where a dependent variable is related to a set of independent or explanatory variables. In all these regression models, y is the 'dependent' or 'explained' variable and $x_1, x_2, \dots$ are the 'independent' or 'explanatory' variables. Thus y depends upon x , but the reverse is not true. In real life, however, we normally find variables that are dependent on each other. In this sort of relationship, the equilibrium values of the variables can be determined by considering the simultaneity in relationship.

In simultaneous relationships a single equation under investigation is part of a wider phenomenon. In such cases the variables can be explained with the help of a system of equations. Let us take an example. At the macro level, as you know from macroeconomics course, aggregate consumption expenditure depends upon aggregate disposable income. Aggregate disposable income, on the other hand, depends upon the national income and taxes imposed by the government. Moreover, national income is dependent on aggregate consumption expenditure of the economy. Thus, estimation of a single equation between aggregate consumption and disposable income will not provide us with consistent and unbiased estimators.

16.2 SOME EXAMPLES OF SIMULTANEOUS EQUATIONS MODELS

In regression models one of the crucial assumptions is that the exogenous variables x 's are independent of the error term u (see Unit 4). This assumption is violated in simultaneous relationships. Let us illustrate this through the demand and supply model.

We can write the demand function as

$$q = \alpha + \beta p + u \quad \dots(16.1)$$

where q represents the quantity demanded, p the price, and u the stochastic error term, representing random shifts in the demand function.

From microeconomic theory we know that the equilibrium price and quantity are determined in the market by intersection of supply and demand curves (see Fig. 16.1). Therefore, we cannot determine equilibrium price by solving the demand equation independently. Again, if there is a random shift in the demand curve we see that it will affect both equilibrium price and quantity (considering the normal slope of supply curve as in panel-a of Fig. 16.1). In case the supply curve is vertical, then a shift in the demand curve affects equilibrium price only (see panel-b of Fig. 16.1). Again, if the supply curve is horizontal, a shift the demand curve affects equilibrium quantity only keeping the level of equilibrium price unchanged (see panel-c of Fig. 16.1). Thus we see that demand curve alone cannot determine the equilibrium price and

quantity. Moreover, the slope of the supply curve influences the relationship between price and quantity. If we consider the demand equation only, the slope of the supply curve gets included in the term ' u '. The inclusion of the slope of supply curve in the error term leads to correlation between price and error term, which violates one of the basic assumptions of OLS, that is, $cov(Xu) = 0$. So, if we ignore the supply curve and estimate the demand equation (16.1) by OLS, then it will produce inconsistent estimates of the parameters.

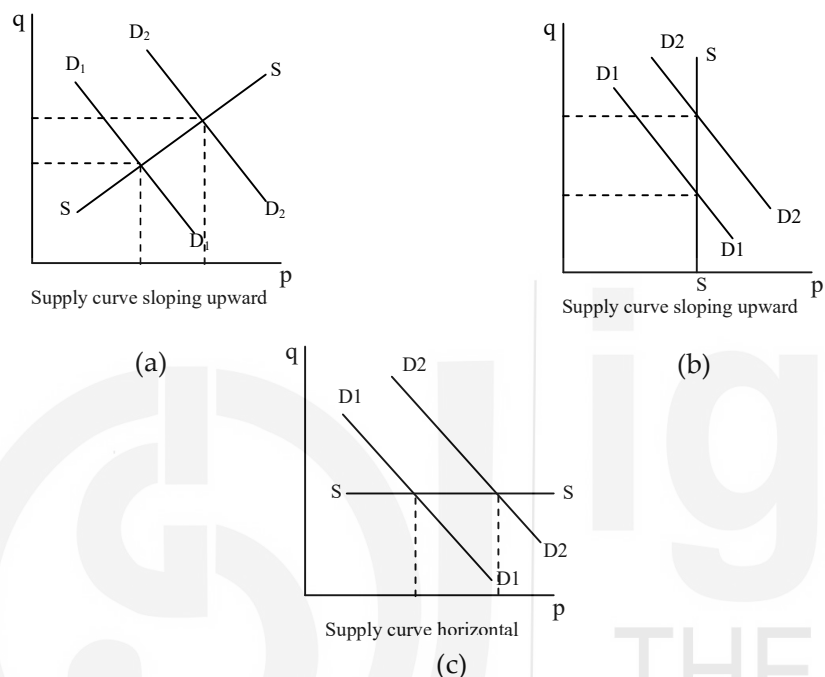


Fig. 16.1: Interaction between Supply and Demand Curves

The problem is that in the demand-supply model we cannot consider the demand function in isolation when we study the relationship between quantity and price as in equation (16.1). The solution is to bring the supply function into picture; and estimate the demand function and the supply function together. Such models are known as simultaneous equations models.

An econometric model is based on certain theoretical concepts. Thus, the simultaneous equations model (SEM) can be best understood by some practical examples. Here we present three examples of SEM.

Example 16.1

Wage-Price Model

The model of money wage and price determination is a good example of SEM. According to the model wage rate is determined by unemployment rate in the economy and the rate of price change (that is, inflation) in an unrestricted labour market (reflected in equation (16.2)). The rate of inflation, on the other hand, is a function of the change in wage rate and money supply

(reflected in equation (16.3)). The model consists of the two equations as follows:

$$W_t = \alpha_0 + \alpha_1 U_t + \alpha_2 P_t + u_{1t} \quad \dots(16.2)$$

$$P_t = \beta_0 + \beta_1 W_t + \beta_2 R_t + \beta_3 M_t + u_{2t} \quad \dots(16.3)$$

where W = rate of change in money wage

U = unemployment rate in percentage

P = rate of change in prices

R = rate of change in cost of capital

M = supply of money

t = time

u_1, u_2 = stochastic disturbances

As the price variable P enters the wage equation (16.2) and the wage variable W enters the price equation (16.3), both the two variables are inter-dependent. Moreover, the explanatory variables P in (16.2) and W in (16.3) are stochastic in nature. Recall that in OLS it is assumed that the explanatory variables are not stochastic. Therefore, these stochastic explanatory variables are expectedly correlated with the stochastic disturbance terms. This correlation violates the assumption of OLS that error term is independent of independent variable. Thus, the classical OLS method is not applicable for estimation of the parameters of the two equations individually.

Example 16.2

Keynesian Model of Income Determination

Let us consider the simple Keynesian model of income determination:

$$\text{Consumption function } C_t = \beta_0 + \beta_1 Y_t + u_t \quad 0 < \beta_1 < 1 \quad \dots (16.4)$$

$$\text{Income identity: } Y_t = C_t + I_t (= S_t) \quad \dots (16.5)$$

where C = consumption expenditure

Y = income

I = Investment (assumed to be exogenous)

S = Savings

t = time

u = stochastic disturbance term

β_0 and β_1 are parameters.

The parameter β_1 represents the marginal propensity to consume (MPC). From macroeconomic theory we know that β_1 is expected to remain between 0 and 1. Equation (16.4) is the consumption function which shows that consumption consists of two parts: autonomous consumption expenditure (β_0) and income induced consumption ($\beta_1 Y$). Equation (16.5) is the national income identity, signifying that total income is equal to total consumption expenditure plus total investment expenditure.

From the consumption function (16.4) we find that C and Y are interdependent and that Y_t is not expected to be independent of the disturbance term, u_t . When u_t changes (because of a variety of factors included in the error term) the consumption function shifts, which, in turn affects Y_t . Therefore, once again the classical least-squares method is not applicable to (16.4). If applied, the estimators thus obtained will be inconsistent.

Example 16.3

Macroeconomic Model

The celebrated IS model, or goods market equilibrium model of macroeconomics (see Unit 3 of MEC-102) in its non-stochastic form can be expressed as

$$\text{Consumption function: } C_t = \beta_0 + \beta_1 Y_{dt} \quad 0 < \beta_1 < 1 \quad \dots(16.6)$$

$$\text{Tax function: } T_t = \gamma_0 + \gamma_1 Y_t \quad 0 < \gamma_1 < 1 \quad \dots(16.7)$$

$$\text{Investment function: } I_t = \theta_0 + \theta_1 r_t \quad \dots(16.8)$$

$$\text{Disposable income: } Y_{dt} = Y_t - T_t \quad \dots(16.9)$$

$$\text{Government expenditure: } G_t = \bar{G} \quad \dots(16.10)$$

$$\text{National income identity: } Y_t = C_t + I_t + G_t \quad \dots(16.11)$$

where Y = national income

C = consumption spending

I = planned or desired net investment

G = Given level of government expenditure

T = taxes

Y_d = disposable income

r = interest rate

In order to solve the above equation system, we first substitute (16.7) into (16.9); and then the value of Y_{dt} into (16.6). Subsequently, we put the resulting

equation for C , and equation (16.8) for I and (16.10) for G into (16.11). Consequently, we obtain

$$Y_t = [\beta_0 + \beta_1(Y_t - T_t)] + [\theta_0 + \theta_1 r_t] + \bar{G}$$

By re-arranging terms in the above equation, we obtain

$$Y_t = \pi_0 + \pi_1 r_t \quad \dots(16.12)$$

where

$$\pi_0 = \frac{\beta_0 - \alpha_0 \beta_1 + y_0 + \bar{G}}{1 - \beta_1(1 - \alpha_1)}$$

and

$$\pi_1 = \frac{1}{1 - \beta_1(1 - \alpha_1)}$$

Equation (16.12) is the equation of the IS curve, which provides the combinations of the interest rate and level of income such that the goods market is in equilibrium.

What would happen if we were to estimate, say, the consumption function (16.6) in isolation? Could we obtain unbiased and/or consistent estimates of β_0 and β_1 ? Such a result is unlikely because consumption depends on disposable income, which depends on national income Y . But Y depends on r and G as well as the other parameters entering in π_1 . Therefore, unless we consider all these influences, a simple regression of C on Y_d is bound to give biased and/or inconsistent estimates of β_0 and β_1 .

16.3 ENDOGENOUS VARIABLES AND EXOGENOUS VARIABLES

There are two types of variables in simultaneous equations models: endogenous and pre-determined variables.

- (i) *Endogenous Variables* are determined jointly and interdependently within the model. The endogenous variables are determined by the exogenous variables. We use notation Y for endogenous variables.
- (ii) *Predetermined Variables*: Predetermined variables are of three types: (i) exogenous variables, (ii) lagged exogenous variables, and (iii) lagged endogenous variables. Exogenous variables are determined outside the equation system. In other words, these variables are given from outside. For example, in the production function of wheat, rainfall is an exogenous variable.

Remember that exogenous variables determine endogenous variables, but endogenous variables do not determine exogenous variables.

In a simultaneous equations model, we can include the lagged values of endogenous variables. For example, C_t is current year consumption and is assumed to be an endogenous variable. Consumption in the previous time period (C_{t-1}) is a lagged endogenous variable. Similarly, we can include lagged exogenous variables (for example, X_{t-1}) as an explanatory variable in an equation of the simultaneous equations system.

Notations

To facilitate our discussion, we introduce the following example: Let us consider a SEM with M endogenous and K exogenous variables. The general M equations model in M endogenous, or jointly dependent, variables may be written as Equation (16.13) which is a time dependent SEM:

$$Y_{1t} = \beta_{12}Y_{2t} + \beta_{13}Y_{3t} + \dots + \beta_{1M}Y_{Mt} + \gamma_{11}X_{1t} + \gamma_{12}X_{2t} + \dots + \gamma_{1K}X_{Kt} + u_{1t}$$

$$Y_{2t} = \beta_{21}Y_{1t} + \beta_{23}Y_{3t} + \dots + \beta_{2M}Y_{Mt} + \gamma_{21}X_{1t} + \gamma_{22}X_{2t} + \dots + \gamma_{2K}X_{Kt} + u_{2t}$$

$$Y_{3t} = \beta_{31}Y_{1t} + \beta_{32}Y_{2t} + \dots + \beta_{3M}Y_{Mt} + \gamma_{31}X_{1t} + \gamma_{32}X_{2t} + \dots + \gamma_{3K}X_{Kt} + u_{3t}$$

.....

$$Y_{Mt} = \beta_{M1}Y_{1t} + \beta_{M2}Y_{2t} + \dots + \beta_{M,M-1}Y_{M-1,t} + \gamma_{M1}X_{1t} + \gamma_{M2}X_{2t} + \dots + \gamma_{MK}X_{Kt} + u_{Mt}$$

...(16.13)

where $Y_1, Y_2, \dots, Y_M$ are M endogenous variables

$X_1, X_2, \dots, X_K$ are K predetermined variables (one of these X variables may take a value of unity to allow for the intercept term in each equation)

$u_1, u_2, \dots, u_M$ are M stochastic disturbances

$t = 1, \dots, T$ are Total number of observations

β 's are coefficients of the endogenous variables

γ 's are coefficients of the predetermined variables

As equation (16.13) shows, there are three types of variables: First, we have endogenous variables, whose values are determined within the model. Second, we have predetermined variables, whose values are determined outside the model. Third, we have the stochastic error terms. The endogenous variables are regarded as stochastic whereas the predetermined variables are non-stochastic in nature.

In notations, X_{1t} is a current (present-time) exogenous variable, whereas $X_{1(t-1)}$ is a lagged exogenous variable. Note that $Y_{(t-1)}$ is a lagged

endogenous variable with a lag of one time period. Since the value of $Y_{1(t-1)}$ is known at the current time t , it is regarded as non-stochastic. Hence, it is considered as a predetermined variable¹. In short, current exogenous, lagged exogenous, and lagged endogenous variables are considered as predetermined, as their values are not determined by the model in the current time period.

At this point a question may be taking shape in your mind, how do we construct econometric models? What are the variables that should be included in an equation? Two issues are very important. First, such questions are answered by economic theory. The variables in an econometric model are included based on certain economic theory. In some cases, which are not established theory yet, the equations depend upon the logic that the researcher puts forth. Thus, behind the inclusion of a variable in an equation, there is some logic. Second, it is up to the concepts of economic theory and the model builder to specify the variables that are endogenous and that are predetermined. Although non-economic variables (variables not determined by any economic theory), such as temperature and rainfall, are clearly exogenous or predetermined, the model building must be exercised with great care while classifying economic variables as endogenous or predetermined. One must be able to defend i) the classification of variables into endogenous and exogenous, and ii) inclusion of a particular variable in an equation.

Check Your Progress 1

1) Explain the meaning of each of the following terms:

- Endogenous Variables.
- Exogenous Variables

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2) Distinguish between pre-determined variable and exogenous variable.

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3) Write the general form of a simultaneous equation model.

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¹It is assumed implicitly here that the stochastic disturbances, the u 's, are serially uncorrelated. If this is not the case, Y_{t-1} will be correlated with the current period disturbance term u_t . Hence, we cannot treat it as pre-determined.

16.4 SIMULTANEITY BIAS

We have mentioned earlier that in simultaneous models the explanatory variables and the error terms are not independent. An important assumption of classical regression model is that there is no covariance between the explanatory variable and the error term.

Let us look into the simple Keynesian model of income determination given in Example 16.2.

$$\text{Consumption function } C_t = \beta_0 + \beta_1 Y_t + u_t \quad 0 < \beta_1 < 1 \quad \dots (16.4)$$

$$\text{Income identity: } Y_t = C_t + I_t \dots (16.5)$$

Suppose we want to estimate the parameters of the consumption (16.4) without considering the income identity (16.5). The standard assumptions under classical regression model for equation (16.4) will be

$$E(u_t) = 0,$$

$$E(u_t)^2 = \sigma^2,$$

$$E(u_t u_{t+i}) = 0 \text{ for } i \neq 0$$

and

$$\text{cov}(Y_t u_t) = 0$$

Note that in this case we take C as the dependent variable and Y as the independent variable. The assumption $\text{cov}(Y_t u_t) = 0$ gets violated in this case, as Y is not independent of u . Let us look into the consequences.

16.4.1 Endogeneity Problem

In the case of the consumption function (16.4) we will first show that Y_t and u_t are correlated and then prove that OLS estimator $\widehat{\beta}_1$ is biased and inconsistent.

To prove that Y_t and u_t are correlated, we substitute (16.4) into (16.5) to obtain the following equation.

$$Y_t = (C_t + I_t) = (\beta_0 + \beta_1 Y_t + u_t) + I_t$$

or,

$$Y_t - \beta_1 Y_t = \beta_0 + u_t + I_t$$

or,

$$(1 - \beta_1)Y_t = \beta_0 + u_t + I_t$$

Therefore,

$$Y_t = \frac{\beta_0}{1-\beta_1} + \frac{1}{1-\beta_1}I_t + \frac{1}{1-\beta_1}u_t \quad \dots(16.14)$$

$$E(Y_t) = \frac{\beta_0}{1-\beta_1} + \frac{1}{1-\beta_1}I_t + \frac{1}{1-\beta_1}E(u_t)$$

or,

$$E(Y_t) = \frac{\beta_0}{1-\beta_1} + \frac{1}{1-\beta_1}I_t \quad \dots (16.15)$$

Subtracting (16.15) from (16.14) results in

$$[Y_t - E(Y_t)] = \frac{1}{1-\beta_1}u_t \quad \dots(16.16)$$

We have assumed that $E(u_t) = 0$. Therefore,

$$u_t = [u_t - E(u_t)] \quad \dots(16.17)$$

The covariance between Y_t and u_t is given by

$$\begin{aligned} cov(Y_t u_t) &= E[y_t u_t] \\ &= E[Y_t - E(Y_t)][u_t - E(u_t)] \end{aligned}$$

Since $[Y_t - E(Y_t)] = \frac{1}{1-\beta_1}u_t$ from (16.16), we have

$$\begin{aligned} cov(Y_t u_t) &= E\left[\frac{1}{1-\beta_1}u_t u_t\right] \\ &= \frac{E(u_t)^2}{1-\beta_1} = \frac{\sigma^2}{1-\beta_1} \quad \dots(16.18) \end{aligned}$$

Since σ^2 is positive by definition, the covariance between Y and u given in (16.18) is bound to be different from zero. As a result, the explanatory variable Y_t and the error term u_t in equation (16.4) are correlated, which violates the assumption of the classical linear regression model that the disturbances are independent of (or not uncorrelated with) the explanatory variables.

In simple language, C determines Y (equation (16.4)), and Y determines C (equation (16.5)). Therefore, we must estimate both the equations together.

The OLS estimator is biased because of correlation between Y_t and u_t . we proceed as follows:

The OLS estimator of β_1 is given by

$$\hat{\beta}_1 = \frac{\sum y_t c_t}{\sum y_t^2} \quad \dots (16.19)$$

In deviation form, equation (16.4) can be expressed as

$$c_t = \beta_1 y_t + u_t$$

By substituting the above equation in (16.19) we obtain

$$\hat{\beta}_1 = \frac{\sum y_t c_t}{\sum y_t^2} = \frac{\sum y_t (\beta_1 y_t + u_t)}{\sum y_t^2} = \beta_1 + \frac{\sum y_t u_t}{\sum y_t^2} \quad \dots (16.20)$$

$$E(\hat{\beta}_1) = \beta_1 + \frac{\sum E(y_t u_t)}{\sum y_t^2}$$

or,

$$E(\hat{\beta}_1) - \beta_1 = \frac{\sum E(y_t u_t)}{\sum y_t^2} \quad \dots (16.21)$$

Equation (16.21) gives the extent of simultaneity bias in the present case. As the expectation-operator is a linear operator, we cannot evaluate $E\left[\frac{\sum y_t u_t}{\sum y_t^2}\right]$, since $E\left(\frac{A}{B}\right) \neq \frac{E(A)}{E(B)}$. But intuitively it is clear that $\hat{\beta}_1$ will be an unbiased estimator of β_1 only if $\left[\frac{\sum y_t u_t}{\sum y_t^2}\right] = 0$.

16.4.3 The OLS Estimator is not Consistent

Though we have shown in (16.18) that the covariance between Y and u is not zero, it does not mean $\text{cov}(y_t u_t) \neq 0$, since $\text{cov}(y_t u_t)$ is a population concept. It is not exactly equal to $\sum y_t u_t$ which is a sample measure, although as the sample size increases indefinitely the latter will tend toward the former. But if the sample size increases indefinitely, then we can resort to the concept of consistent estimator and thus find out what happens to β_1 as n , the sample size, increases indefinitely.

Now an estimator is said to be consistent if its **probability limit**, or **plim** for short, (see Unit 3 for definition) is equal to its true (population) value. Therefore, to show that $\hat{\beta}_1$ of (16.20) is inconsistent; we must show that its *plim* is not equal to the true β_1 .

Applying the rules of probability limit (*plim*) to (16.20), we obtain¹

$$\begin{aligned}\text{plim}(\hat{\beta}_1) &= \text{plim}(\beta_1) + \text{plim}\left(\frac{\sum y_t u_t}{\sum y_t^2}\right) \\ &= \text{plim}(\beta_1) + \text{plim}\left(\frac{\sum y_t u_t / n}{\sum y_t^2 / n}\right) \\ &= \beta_1 + \frac{\text{plim}(\sum y_t u_t / n)}{\text{plim}(\sum y_t^2 / n)} \quad \dots(16.22)\end{aligned}$$

In the second step of equation (16.22) we have divided $\sum y_t u_t$ and $\sum y_t^2$ by the sample size, n , so that the quantities in the parentheses are now the sample covariance between Y and u , and the sample variance of Y respectively.

In other words, (16.22) states that the *plim* of $\hat{\beta}_1$ is equal to true β_1 plus the ratio of two quantities: (i) the *plim* of the sample covariance between Y and u , and (ii) the *plim* of the sample variance of Y . Now, as the sample size n increases indefinitely, we expect the sample covariance between Y and u to approximate the population covariance $E[Y_t - E(Y_t)][u_t - E(u_t)]$. From (16.18) we know that

$$\text{cov}(Y_t u_t) = [\sigma^2 / (1 - \beta_1)]$$

Similarly, as n tends to infinity, the sample variance of Y will approximate its population variance, say σ_y^2 . Therefore, equation (16.22) may be written as

$$\begin{aligned}p \lim(\hat{\beta}_1) &= \beta_1 + \frac{\sigma^2 / (1 - \beta_1)}{\sigma_y^2} \\ &= \beta_1 + \frac{1}{(1 - \beta_1)} \left(\frac{\sigma^2}{\sigma_y^2} \right) \quad \dots(16.23)\end{aligned}$$

It is assumed that $0 < \beta_1 < 1$. Further, σ^2 and σ_y^2 are both positive. Thus, equation (16.23) implies that *plim* ($\hat{\beta}_1$) will always be greater than β_1 ; that is, $\hat{\beta}_1$ will overestimate the true parameter², β_1 . In other words, $\hat{\beta}_1$ is a biased estimator, and the bias will not disappear; no matter how large the sample size is.

16.4.2 Solution to Simultaneity Bias

In the preceding section we learnt that if we apply OLS to an equation belonging to a system of simultaneous equations, we will not get unbiased and consistent estimators. To obtain unbiased and consistent estimators, we apply certain other methods of estimation. There are several methods for this purpose. The most common methods are:

¹As mentioned in Unit 1, the *plim* of a constant (for example, β_1) is the same constant and the *plim* of $(A/B) = \text{plim}(A)/\text{plim}(B)$. Note, however, that $E(A/B) \neq E(A)/E(B)$.

²In general, however, the direction of the bias will depend on the structure of the model and the parameters.

- (i) Reduced form method, or indirect least squares (ILS)
- (ii) Instrumental variables (IV)
- (iii) Two-stage least squares (2SLS)
- (iv) Limited information maximum likelihood (LIML)
- (v) Three stage least squares (3SLS)
- (vi) Full information maximum likelihood (FIML)

Some of these methods will be discussed in the course 'MECE 102: Advanced Econometric Methods'.

16.5 STRUCTURAL FORM AND REDUCED FORM

We presented the simultaneous equations system in (16.13). These equations are known as the 'structural equations' or 'behavioral equations' as they represent the structure (of an economic model) or the behavior of an economic agent (e.g., consumer or producer). The β 's and γ 's in (16.13) are known as the structural parameters. Recall that in the structural equations both endogenous and exogenous variables are on the right-hand side of each equation.

By solving for the M endogenous variables from the structural equations we can derive the reduced-form equations and the associated reduced-form coefficients. A reduced-form equation is one where each equation contains one endogenous variable and that endogenous variable is expressed solely in terms of the predetermined variables and the stochastic disturbances. To illustrate, let us consider the Keynesian model of income determination given at Example 16.2 again:

$$\text{Consumption function: } C_t = \beta_0 + \beta_1 Y_t + u_t \quad \dots(16.24)$$

$$\text{where } 0 < \beta_1 < 1$$

$$\text{Income identity: } Y_t = C_t + I_t \quad \dots(16.25)$$

In the above model C (consumption) and Y (income) are endogenous variables and I (investment expenditure) is an exogenous variable. While the first equation is stochastic in nature, the second one is an identity. As usual, the MPC (β_1) is assumed to remain between 0 and 1. Equations (16.24) and (16.25) are called the 'structural form' of the model. Here, β_0 and β_1 are structural parameters of the model.

If we substitute (16.24) in (16.25), we obtain the following:

$$Y_t = C_t + I_t = [\beta_0 + \beta_1 Y_t + u_t] + I_t$$

By re-arranging terms in the above equation, we obtain

$$\begin{aligned} Y_t - \beta_1 Y_t &= \beta_0 + I_t + u_t \\ \text{or,} \\ (1 - \beta_1) Y_t &= \beta_0 + I_t + u_t \end{aligned}$$

or,

$$Y_t = \frac{\beta_0}{(1 - \beta_1)} + \frac{1}{(1 - \beta_1)} I_t + \frac{1}{(1 - \beta_1)} u_t$$

The above can be represented as

$$Y_t = \Pi_0 + \Pi_1 I_t + w_t \quad \dots(16.26)$$

where $\Pi_0 = \frac{\beta_0}{1 - \beta_1}$

$$\Pi_1 = \frac{1}{1 - \beta_1}$$

$$w_t = \frac{u_t}{1 - \beta_1}$$

Equation (16.26) is a 'reduced form' equation. It expresses the endogenous variable Y solely as a function of the exogenous (or pre-determined) variable and the stochastic disturbance term. Π_0 and Π_1 are 'reduced-form parameters'. Notice that these reduced-form parameters are nonlinear combinations of the structural parameters.

Substituting the value of Y_t from (16.26) into C of (16.24), we obtain another reduced-form equation:

$$C_t = \Pi_2 + \Pi_3 I_t + w_t \quad \dots(16.27)$$

where $\Pi_0 = \frac{\beta_0}{1 - \beta_1}, \Pi_3 = \frac{\beta_0}{1 - \beta_1}, \text{ and } w_t = \frac{u_t}{1 - \beta_1}$

The reduced-form coefficients, such as Π_1 and Π_2 and Π_3 , are also known as impact, or short-run multipliers, because they measure the immediate impact on the endogenous variable of a unit change in the value of the exogenous variable. If in the preceding Keynesian model the investment expenditure is increased by, say, Re.1 and if the MPC is assumed to be 0.8, then from (16.25) we obtain $\Pi_1 = 5$. This result means that increasing the investment by Re.1 will immediately (i.e., in the current time period) lead to an increase in income of Rs. 5, that is, a fivefold increase. Similarly, under the assumed conditions, equation (16.25) shows that $\Pi_3 = 4$, meaning that Re. 1 increase in investment expenditure will lead immediately to an increase of Rs. 4 in consumption expenditure.

In the context of econometric models, equations such as (16.25) or $Q_{dt} = Q_{st}$ (Quantity demanded is equal to quantity supplied) are known as equilibrium condition. On the other hand, identity states that aggregate income Y must be equal to aggregate expenditure (i.e., consumption expenditure plus investment

expenditure). When equilibrium is achieved, the endogenous variables assume their equilibrium values³.

16.6 CONCEPT OF IDENTIFICATION

In the previous section we presented the reduced form of an equation. In the reduced form an endogenous variable is explained in terms of the pre-determined variables in the simultaneous equations system and the stochastic error term. Thus, we can apply OLS to the reduced form and obtain estimators of the reduced form parameters.

Our objective, however, is to find out the structural form parameters. The concept of identification is used to denote the possibility (or impossibility) of deducing the values of the structural parameters from the reduced form parameters.

The identification issue arises prior to estimation of the model. If more than one theory is consistent with the same data (we illustrate this in Example 16.4 below), then they are said to be observationally equivalent, and we cannot distinguish them. In such cases the structure of the econometric model is said to be unidentified and there is a need to modify the model.

Identification is a problem of model formulation, rather than of model estimation or model appraisal. A model is said to be 'identified' if it is in a unique statistical form, where we can obtain unique estimator for each of its structural parameters. If a model is 'not identified', then we cannot say exactly what relationship we are estimating.

Example 16.4

Identification Problem in the Demand-Supply Model

To measure the coefficients of the demand equation, normally the published time series reporting the quantity bought of the commodity is used. However, the quantity bought is identical with the quantity sold at the given price. Market data register points of interaction of equilibrium supply and demand at the price prevailing in the market at a certain point of time. A sample of time-series observations shows simultaneously the quantity demanded, and the quantity supplied, at the prevailing market price. That is, it only shows the points of interactions of demand and supply. If we use these data for estimation, we measure the coefficients of a function of the form $Q = f(p)$. This equation may be either the demand function or the supply function. How can we be sure whether this equation represents demand function or supply function? If person A is interested in estimation of the demand function, then (s)he can use the market data. Similarly, if person B is interested in estimation of the supply equation, (s)he will also be using the same market data! Obviously, we need some criteria, which will enable us to verify that the estimated coefficients belong to a particular relationship. Such criteria are known as 'identification conditions' of a function. We will explain these criteria in Section 16.6.

³ For details see Jan Kmenta (1997), *Elements of Econometrics*, 2nd Edition, Macmillan, New York, pp. 723-731.

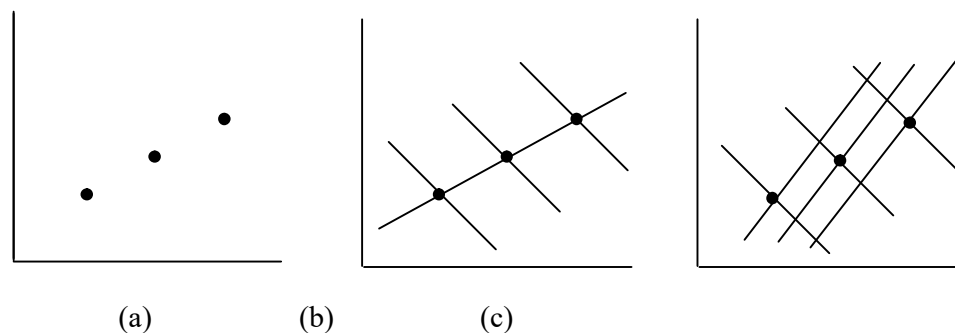


Fig. 16.2: Supply and Demand Functions

In Fig. 16.2 we plot market data for three observations. Fig. 16.2(a) shows three data points. We have no knowledge of the conditions of supply and demand beyond our consideration that the data represents the equilibrium points, that is, the equality between supply and demand. In Fig. 16.2(b) we try to introduce supply and demand functions for these three observations. Notice that the supply curve is stable in Fig. 16.1(b) whereas the demand curve is shifting. Thus, the data represents points on the supply equation under the assumption that the supply curve is stable and the demand curve shifts. In Fig. 16.2(c), on the other hand, we show that both the demand curve and the supply curve are shifting randomly. Thus, if we wish to estimate $Q = f(p)$ from the same data set, then we estimate neither the demand curve nor the supply curve; rather we end up estimating the mongrel equation⁴.

16.6.1 Paradox of Identification

Discussion in the preceding section suggests that if we want to measure a given function belonging to a simultaneous-equations model, the function must be fairly stable over the sample period, that is, it must shift within a smaller range as compared with other relationships of the same model. In Example 16.4 we have shown that we can measure the supply function when it is fairly stable and the demand function is shifting. Similarly, we can measure the demand function if it is fairly stable while the supply function shows adequate variability.

It can happen when the factor causing a shift in a function is absent in another function. In other words, to identify the supply function (see Fig. 16.2(b)), some factors absent from it but included in the demand function. So, to identify an equation in a simultaneous equations model, we have to look for the variables absent from it (but present in other equations of the model). This is called the paradox of identification. In identification problem, the relevant variables are the ones that are absent from the equation!

Example 16.5

Let us consider the supply-demand equation system. We write the system in equilibrium form (that is, with $q_s = q_d = q$) as

$$q_s = \alpha_1 p + \beta_1 z_1 + u_1 \quad \dots(16.28)$$

and

⁴ A combination of two and more equations is called *amongrel* equation.

$$q_d = \alpha_2 p + u_2 \quad \dots (16.29)$$

To be more precise, let q be per capita rice consumption at the country level, let p be the average price per kg of rice in the county, and let z_1 be the price of fertilizer. While q and p are assumed to be endogenous, z_1 is assumed to be exogenous. It means that (16.28) must be the supply function, as the price of fertilizer would shift supply ($\beta_1 < 0$) but not demand. The demand function contains no exogenous variables.

Intuitively, we can say that the demand equation (16.29) is identified. We have an observed variable z_1 that shifts the supply equation; but it does not affect the demand equation. Thus, the demand function can be traced by taking different values of z_1 .

The supply equation (16.28), on the other hand, cannot be traced out because there is no exogenous observed factor that leads to a shift in the demand curve. Only the stochastic error term (which is not observable) leads to a shift in the demand equation. May be in the presence of certain factors (such as preference of people towards wheator increase in per capita income) we can identify the supply equation.

To summarize: in the system of (16.28) and (16.29), it is the presence of an exogenous variable (z_1) in the supply equation that allows us to estimate the demand equation. Further, we have to check the identification status of separately for each equation in the model.

Let us consider a simultaneous equations model with two equations.

$$y_1 = \alpha_1 + \beta_1 y_2 + z_1 \beta_2 + u_1 \quad \dots (16.30)$$

and

$$y_2 = \alpha_2 + \beta_3 y_1 + z_2 \beta_4 + u_2 \quad \dots (16.31)$$

where y_1 and y_2 are the endogenous variables, and u_1 and u_2 are the error terms. The intercept in the first equation is α_1 , and the intercept in the second equation is α_2 . The fact that z_1 and z_2 are different exogenous variables means that we have imposed exclusion restrictions on the model. In other words, we assume that certain exogenous variables do not appear in the first equation while some exogenous variables do not appear in the second equation. The key question is: Under what conditions can we estimate the parameters in, say, (16.31)? This is the identification issue.

16.6.2 Identification Status of an Equation

As mentioned earlier, identification refers to the possibility of finding out estimators of all the parameters of an equation. Thus, we have to move equation-wise. There are two possibilities: whether the equation under consideration is identified or not identified.

1. Equation is under-identified

If the structure of the equation is not unique or there is not sufficient information to uniquely distinguish the equation, then the equation is under-identified.

2. Equation is identified

- (a) Exactly identified: If there is sufficient information available to distinguish the equation then it is called just identified or exactly identified equation
- (b) Over-identified: If there is more than sufficient information regarding the equation then the equation is called over-identified.

Identification is closely related to the estimation of the model.

- (1) If an equation is identified, its coefficient can, in general, be statistically estimated. In particular:
 - (a) If the equation is exactly identified, the appropriate method to be used for its estimation is the method of indirect least squares (ILS).
 - (b) If the equation is over-identified, indirect least squares cannot be applied, because it will not yield unique estimates of the structural parameters. There are various methods which can be used in this case, for example two-stage least squares (2SLS), or maximum likelihood methods. These methods will be developed in Unit 15.

Identification may be established either by the examination of the specification of the structural model, or by the examination of the reduced form of the model. Reduced form approach is conceptually confusing and computationally more difficult than the structural model approach, because it requires the derivation of the reduced form first and then examination of the values of the determinant formed from some of the reduced form coefficients. The structural form approach is simpler and more useful. In this unit we will only discuss the structural form approach of identification.

In applying the identification rules, we should ignore the intercept term if it is present in each equation. If intercept term is present in some equations of the model, not in others, then we should treat it like a separate variable (say X_0), that takes the value 1 for presence of the intercept term, and 0 otherwise.

Check Your Progress 2

- 1) Explain the following concepts:

- Structural equations
- Reduced Form Equations
- Simultaneity Bias

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- 2) Why is it called the paradox of identification?

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- 3) Explain, with the help of an example, the concept of simultaneity bias.

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16.7 IDENTIFICATION CONDITIONS

There are two conditions that must be fulfilled for an equation to be identified.

16.7.1 Order Condition of Identification

This condition is based on a counting rule of the variables included and excluded from a particular equation. It is a necessary but not sufficient condition for identification of an equation. The order condition may be stated as follows:

For an equation to be identified the total number of variables (endogenous and exogenous) excluded from it must be equal to or greater than the number of endogenous variables in the model less one. Remember that in a simultaneous equation system, the number of endogenous variables is equal to the number of equations of the model. Thus, the order condition can be stated as follow:

For an equation to be identified the total number of variables excluded from it but included in other equations must be at least as large as the number of equations of the system less one.

Suppose,

G = Total number of equations (=total number of endogenous variables)

K = Total number of variables in the model (endogenous and predetermined)

M = Total number of variables, endogenous and exogenous, included in a particular equation

Then the order condition for identification may be symbolically expressed as

$$(K-M) \geq (G-1)$$

$$[\text{Excluded variables}] \geq [\text{Total number of equations} - 1]$$

Example 16.6

Suppose a simultaneous model contains 10 equations with 15 variables (10 endogenous and 5 exogenous).

(a) Let us assume, for the first equation we have

$G = 10$	$K = 15 \quad M = 11$
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Recall that the **order condition** is given by

$$(K - M) \geq (G - 1)$$

In this case, we have

$$(15 - 11) < (10 - 1)$$

Thus, the order condition is not satisfied, and the first equation is under-identified.

(b) Let us assume, for the second equation we have

$G = 10$	$K = 15 \quad M = 5$
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Order condition:

$$(K - M) \geq (G - 1)$$

$$(15 - 5) > (10 - 1)$$

Thus, the order condition is satisfied for the second equation.

The order condition for identification is necessary for a relation to be identified, but it is not sufficient. Therefore, order condition may be fulfilled in an equation and yet the relation may not be identified.

16.7.2 Rank Condition of Identification

The rank condition states that: in a system of G equations an equation is identified if and only if it is possible to construct at least one non-zero determinant of order $(G-1)$ from the coefficients of the variables excluded from that equation but contained in other equations of the model.

This condition is called the rank condition because it refers to the rank of the matrix of parameters of excluded variables. The rank of a matrix is the order of the largest non-zero determinant which can be formed from the matrix. In our case the relevant matrix is the sub-matrix of coefficients of the excluded variables. Hence, the rank condition may be also stated as follows: a sufficient condition for identification of a relationship is that the rank of the matrix of parameters of all the excluded variables (endogenous and predetermined) from that equation be equal to $(G-1)$

The practical steps for tracing the identifiability of an equation of a structural model may be outlined as follows:

Step1: Write down the parameters of all the equations of the model in a separate table, noting that the parameter of a variable excluded from an equation is equal to zero.

Example 16.7

Consider the following simultaneous equations model:

$$y_1 = 3y_2 + 2x_1 + x_2 + u_1$$

$$y_2 = y_3 + x_3 + u_2$$

$$y_3 = y_1 - y_2 - 2x_3 + u_3$$

where the y 's are endogenous variables and the x 's are predetermined variables.

The above model can be written in the following form:

$-y_1 + 3y_2 + 0y_3 - 2x_1 + x_2 + 0x_3 + u_1 = 0$ $0y_1 + y_2 + y_3 - 0x_1 + 0x_2 + x_3 + u_2 = 0$ $y_1 + y_2 + y_3 - 0x_1 + 0x_2 + 2x_3 + u_3 = 0$
--

Ignoring the random disturbances the table of the parameters of the model is as follows:

Table 16.1: Parameters of the Equation System

Equations	Variables					
	y_1	y_2	y_3	x_1	x_2	x_3
1 st equation	-1	3	0	-2	1	0
2 nd equation	0	-1	1	0	0	1
3 rd equation	1	-1	-1	0	0	-2

Step2: Eliminate the row of coefficients of the equation, which is being examined for identification.

For example, if we want to examine the identifiability of the second equation of the model, we strike out the second row of the table of coefficients.

Step3: Eliminate the columns in which a non-zero coefficient of the equation being examined appears. By deleting the relevant row and columns we are left with the coefficients of variables not included in the concerned equation but contained in other equations of the model.

For example, if we are examining for identification the second equation of the system, we will eliminate the second, third and the sixth columns of Table 16.1. Thus, after deleting the required row and columns, we obtain Table 16.2(b).

Table 16.2-(a),(b)

Table of Structural Parameters							Table of parameters left after eliminating the relevant rows and columns		
	y_1	y_2	y_3	x_1	x_2	x_3	y_1	x_1	x_2
1 st	-1	3	0	-2	1	0	-1	-2	1
2 nd	0	-1	1	0	0	1			
3 rd	1	-1	-1	0	0	-2	1	0	0

Step4: Form the determinant(s) of order $(G-1)$ and examine their value. If *atleast one* of these determinants is non-zero, the equation is identified. If all the determinants of order $(G-1)$ are zero, the equation is under-identified.

From Table 16.2(b), we form determinants of order $(G-1) = (3-1)=2$. We obtain three such determinants:

$$\Delta_1 = \begin{vmatrix} -1 & -2 \\ 1 & 0 \end{vmatrix} \neq 0$$

$$\Delta_2 = \begin{vmatrix} -2 & 1 \\ 0 & 0 \end{vmatrix} = 0$$

$$\Delta_3 = \begin{vmatrix} -1 & 1 \\ 1 & 0 \end{vmatrix} \neq 0$$

(the symbol 'Δ' stands for determinant'). We see that we can form two non-zero determinants of order $G-1 = 3-1 = 2$; hence the second equation of our system is identified.

Step5: To see whether the equation is exactly identified or over-identified we use the order condition $(K - M) \geq (G - 1)$. With this criterion, if the equality sign is satisfied, that is, if $(K - M) = (G - 1)$, the equation is exactly identified. If the inequality sign holds, that is, if $(K - M) > (G - 1)$, then the equation is over-identified.

In the case of the second equation, we have

$$G = 3 \quad K = 6 \quad M = 3$$

and the counting rule $(K-M) \geq (G-1)$ gives

$$(6-3) > (3-1)$$

Therefore, the second equation of the model is over-identified. You should check the identification status of the first equation and the third equation. You will find that the first equation is exactly identified according to the order condition and the rank condition. Third, equation fulfils the order condition but does not fulfil the rank condition.

We illustrate the two formal conditions of identification with the help of an example.

Example 16.8

Assume that we have a model describing the market of an agricultural product. From the theory of partial equilibrium, we know that the price in a market is determined by demand and supply. The main determinants of demand are the price of the commodity, the prices of other commodities, incomes, and tastes of consumers. Similarly, important determinants of supply are price of the commodity, other prices, technology, prices of factors of production, and weather condition. The equilibrium condition is that demand be equal to supply. The above theoretical information can be expressed in the form of the following mathematical model:

$$D = a_0 + a_1P_1 + a_2P_2 + a_3Y + a_4t + u$$

$$S = b_0 + b_1P_1 + b_2P_2 + b_3Y + b_4t + w$$

$$D = S$$

where

D = quantity demanded

S = quantity supplied

P_1 = price of the given commodity

P_2 = prices of other commodities

Y = income

C = costs (index of prices of factors of production)

t = time trend. In the demand function it stands for 'taste'; in the supply function it stands for 'technology'

The above model is mathematically complete in the sense that it contains three equations in three endogenous variables: D , S and P_1 . The remaining variables, Y , P_2 , C , t are exogenous. Suppose we want to check the identification status of the supply function. We apply the two criteria for identification:

1. Order condition: $(K-M) \geq (G-1)$

In our example we have

$$K = 7 \qquad M = 5 \qquad G = 3$$

Therefore, $(K-M) = (G-1)$

$$(7-5) = (3-1) = 2$$

Consequently, the second equation satisfies the order condition for identification.

2. Rank condition: The table of the coefficients of the structural model is as follows:

Table 16.3: Coefficients of the Structural Model

Equations	Variables						
	D	P_1	P_2	Y	t	S	C
1 st equation	-1	a_1	a_2	A_2	a_4	0	0
2 nd equation	0	b_1	b_2	0	b_4	-1	b_3
3 rd equation	-1	0	0	0	0	1	0

Following the procedure explained earlier, we eliminate the second row and the second, third, fifth, sixth and seventh columns. Thus, we are left with the table of the coefficients of excluded variables (see Table 16.4 (a) and (b)).

Table 16.4(a): Complete Table of Structural Parameters

-1 $a_1a_2a_3a_4$ 00
0 b_1b_2 0 b_4 1 b_3
-1000010

Table 16.4(b): After eliminating the relevant rows and columns

-1 a_3
-10

The value of the determinant is non-zero, provided that $a_3 \neq 0$.

$$\Delta = \begin{vmatrix} -1 & a_3 \\ -1 & 0 \end{vmatrix} = (0)(-1) - (-1)(a_3) = a_3$$

We see that both the order and rank conditions are satisfied. Hence, the second equation of the model is identified. Furthermore, we see that in the order condition the equality holds: $(7 - 5) = (3 - 1) = 2$. Consequently, the second structural equation is exactly identified.

Thus, we have seen that the order condition of identification is the necessary condition, and the rank condition is 'both necessary and sufficient condition' for identification of an equation. *It implies that even if we do not consider the order condition, we can find out the identification status of an equation through the rank condition.*

The order condition, however, is quite simple to apply. Therefore, we usually begin with the order condition. If the order condition is not fulfilled, we infer that the equation is under-identified. If the order condition is satisfied, we move on to the rank condition.

- 1) Consider the three-equation model

$$Y_1 = \beta_{13}Y_3 + \gamma_{12}X_2 + u_1$$

$$Y_2 = \beta_{21}Y_1 + \beta_{23}Y_3 + \gamma_{21}X_1 + \gamma_{22}X_2 + u_2$$

$$Y_3 = \gamma_{33}X_3 + u_3$$

where Y_1 , Y_2 and Y_3 are endogenous variables, and X_1 , X_2 and X_3 are exogenous variables. Find out the identification status of each equation of the model, based on the order condition and the rank condition.

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- 2) The structure of a model with four endogenous and three exogenous variables is as follows: (1 indicates the presence and 0 indicates the absence of a variable in the equation):

Y_1	Y_2	Y_3	Y_4	X_1	X_2	X_3
1	0	1	1	1	0	0
1	1	1	0	0	1	1
0	0	1	0	1	0	0
1	0	1	1	0	1	0

Which of the four equations is identified?

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16.8 LET US SUM UP

Unlike the single equation model, in simultaneous equations models, we get more than one dependent, or endogenous variable. This endogenous variable may appear in one equation as explained variable and, in another equation, as explanatory variable. Due to the presence of endogenous variables as explanatory variables in some of the equations, the regressors are stochastic in nature. In such situations the classical OLS regression method is not applicable as it may result in inconsistent estimators, and there is simultaneity bias.

Identification of an equation is necessary estimation of the equation. The objective of the identification problem is to obtain unique estimators of the structural parameters from the reduced form parameters. An equation in a

system of simultaneous equations is identified only when we can resolve this issue. If not, the equation remains in unidentified or under-identified situation.

An identified equation can be just-identified or over-identified. In the case of an exactly identified equation, unique estimators of structural parameters can be obtained. In the case of an over identified equation, on the other hand, more than one estimator for certain structural parameters is obtained.

There are two conditions for evaluation of identification status of an equation in a simultaneous equation system, viz., order condition, and rank condition. The order condition is a necessary condition for identification. On the other hand, the rank condition is a necessary and sufficient condition for identification. If the rank condition is satisfied, the order condition is satisfied too, although the reverse is not true.

16.9KEY WORDS

Endogenous Variables	: These are variables which get determined jointly and interdependently within the model.
Exactly-identified Equation	: If there is sufficient information available to distinguish the equation then it is called just identified or exactly identified equation
Exogenous Variables	: These are variables which get determined outside the model, and independently of the endogenous variable.
Identification	: It refers to the possibility (or impossibility) of deducing the values of the parameters.
Inconsistent Estimators	: The value of the estimates may not converge to the population parameter.
Over-identified Equation	: If there is more than sufficient information regarding the equation then the equation is called over-identified.
Paradox of Identification	: It refers to the paradox that identification is done in terms of the variables absent from the equation not in terms of the present variable.
Pre-determined Variables	: The set of exogenous variables and the lagged endogenous variables.
Reduced Form Models	: It refers to models where each equation contains only one dependent variable.
Reduced Form Parameters	: Parameters of the reduced form models.

Simultaneity Bias	: Bias resulting from applying OLS to the simultaneous equations model.
Simultaneous Equations Models	: Models where the equations are related and cannot be solved independently.
Structural Equation	: Equations which represents the structure of an economic model or the behavior of an economic agent.
Structural Parameters	: Parameters appearing in the structural equations.
Under-identified Equation	: If the structure of the equation is not unique or there is not sufficient information to uniquely distinguish the equation then the equation is under-identified.

16.10 SOME USEFUL BOOKS

Gujarati, D. N., D. C. Porter and S. Gunasekar, 2017, *Basic Econometrics*, Fifth Edition, McGraw Hill

Maddala, G. S., and K. Lahiri, 2012, *Introduction to Econometrics*, Fourth Edition, Wiley

Wooldridge, J. M., 2022, *Introductory Econometrics: A Modern Approach*, Seventh Edition, Cengage Learning

16.11 ANSWERS/ HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Refer to Section 16.3 and answer.
- 2) Refer to Section 16.3 and answer. Exogenous variable is one of the categories of pre-determined variables.
- 3) The structure of a simultaneous equations model is given in Section 16.3. Refer to it and answer.

Check Your Progress 2

- 1) Refer to Section 16.5 and answer.
- 2) In order to identify an equation, one has to examine the variables excluded from the equation.
- 3) You should show that OLS estimators are biased and inconsistent. Refer to Section 16.4.

Check Your Progress 3

- 1) First apply the rank condition. Write down the equations as we have done in Example 16.7. Strike out the required row and columns. Check the rank condition.
- 2) Follow the same procedure as in Question 1.

